universität freiburg

Intelligent Embedded Systems Lab

Navigating through Medical, Mobile and Ambient Data

Digital Health

IES Lab
Dr. Mario Cypko
Freiburg, 26. October 2026

Date	Typ	Time	Location	Topics	Lecturer
19.10.2026	L1	12-14	HS 00 036 (G.-Köhler-Allee 101)	Introduction to Digital Health	Prof. Oliver Amft
22.10.2026	E1	16-18	Online	Machine Learning Refresher	Aditya Kumar
26.10.2026	L2	12-14	HS 00 036 (G.-Köhler-Allee 101)	Navigating Through Medical and Ambient Data: Convergence of Mobile Technologies, GDPR, and Contemporary Data Protection Strategies.	Dr. Mario Cypko
29.10.2026	E2	16-18	Online	Medical data: Introduction	Aditya Kumar
02.11.2026	L3	12-14	HS 00 036 (G.-Köhler-Allee 101)	Overview of AI-methods in health applications and medical decision support systems.	Dr. Mario Cypko
05.11.2026	E3	--	--	Medical data: Study week	Aditya Kumar
09.11.2026	L4	12-14	HS 00 036 (G.-Köhler-Allee 101)	Clinical Integration - Medical IT infrastructure with key standards, now and in the future	Dr. Julius Wehrle
12.11.2026	E4	--	--	Medical data: Study week	Aditya Kumar
16.11.2026	L5	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Biomarkers: Timeseries data	Dr. Göran Köber
19.11.2026	E5	16-18	Online	Medical data: Discussion and pitch	Dr. Mario Cypko?
23.11.2026	L6	12-14	Online	Digital Biomarkers: Mental Health	Dr. Lara Puhlmann
26.11.2026	E6	16-18	Online	Temporal data modelling: Introduction	Aditya Kumar
30.11.2026	L7	12-14	HS 00 036 (G.-Köhler-Allee 101)	Human Digital Twin: Concept	Dr. Mario Cypko
03.12.2026	E7	--	--	Temporal data modelling: Study week	Aditya Kumar
07.12.2026	L8	12-14	HS 00 036 (G.-Köhler-Allee 101)	Modeling Digital Twins	Dr. Göran Köber
10.12.2026	E8	--	--	Temporal data modelling: Study week	Aditya Kumar
14.12.2026	L9	12-14	HS 00 036 (G.-Köhler-Allee 101)	Patient Experience Pathway	Dr. Isabel Nieto Alvarez
17.12.2026	E9	16-18	Online	Temporal data modelling: Discussion and pitch	Aditya Kumar
	--	--	--	--	--
11.01.2027	L10	12-14	HS 00 036 (G.-Köhler-Allee 101)	Population-scale digital twins	Dr. Göran Köber
	--	--	--	--	--
18.01.2027	L11	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Patient Twin & Model-Guided Medicine	Dr. Mario Cypko
21.01.2027	E10	16-18	Online	Bayesian Networks: Introduction	Dr. Mario Cypko
25.01.2027	L12	12-14	HS 00 036 (G.-Köhler-Allee 101)	Clinical Perspective on Digital Health	Prof. Michael Quante
28.01.2027	E11	--	--	Bayesian Networks: Study week	Dr. Mario Cypko
01.02.2027	L13	12-14	HS 00 036 (G.-Köhler-Allee 101)	Wisdom-oriented Healthcare System	Prof. Heinz Lemke
TBA	L14	16-18	TBA	Digital Health Research Night	Prof. Oliver Amft
04.02.2027	E12	16-18	Online	Bayesian Networks: Discussion	Dr. Mario Cypko
08.02.2027	L15	12-14	HS 00 036 (G.-Köhler-Allee 101)	Digital Health applications: prescription models, regulations; General Q&A session	Prof. Oliver Amft
TBA	Exam	TBA	TBA	Written exam (60 min)	Prof. Oliver Amft

Intro

15-25 min. Content

Activation

⋮

15-25 min. Content

Activation

Outro

pingo.coactum.de/815896

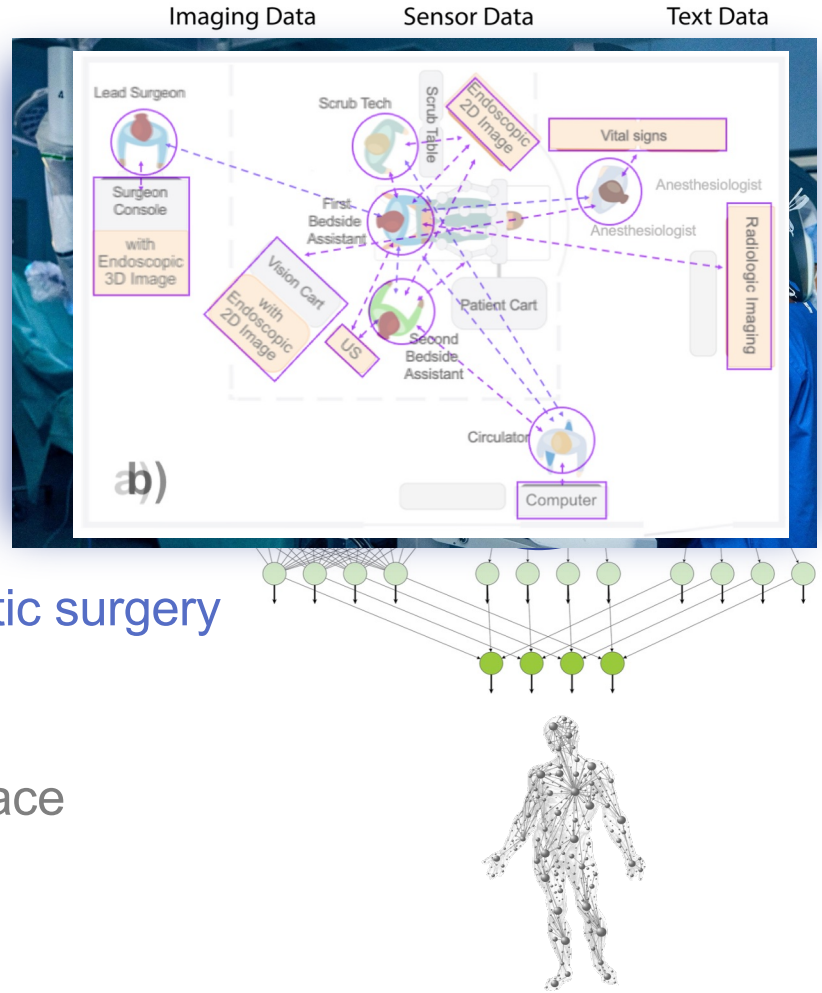
Quick Intro

Computer Science

PhD in CS related to Med AI

Postdocs
* ESA ESTEC / EAC Human Research
* German Heart Center AI for patient triaging
* FU Berlin / Charité HCI in (AI-supported) robotic surgery

Research Group MedAIM
Decision Support for Telehealthcare on Earth & in Space



MedAIM - Medical AI and Modeling Group

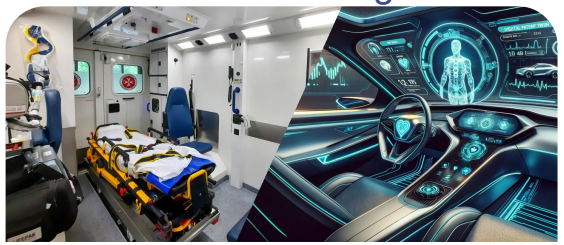
Intelligent Embedded Systems Lab

MedAIM - Medical AI and Modeling Group

Intelligent Embedded Systems Lab

Medical Support in Remote (Extreme) Environments

Continues and Situational - Mobile and Ambient Health Tracking



Primary Patient Care Facility

Highly Specialized Private Clinics

Central City Hospital

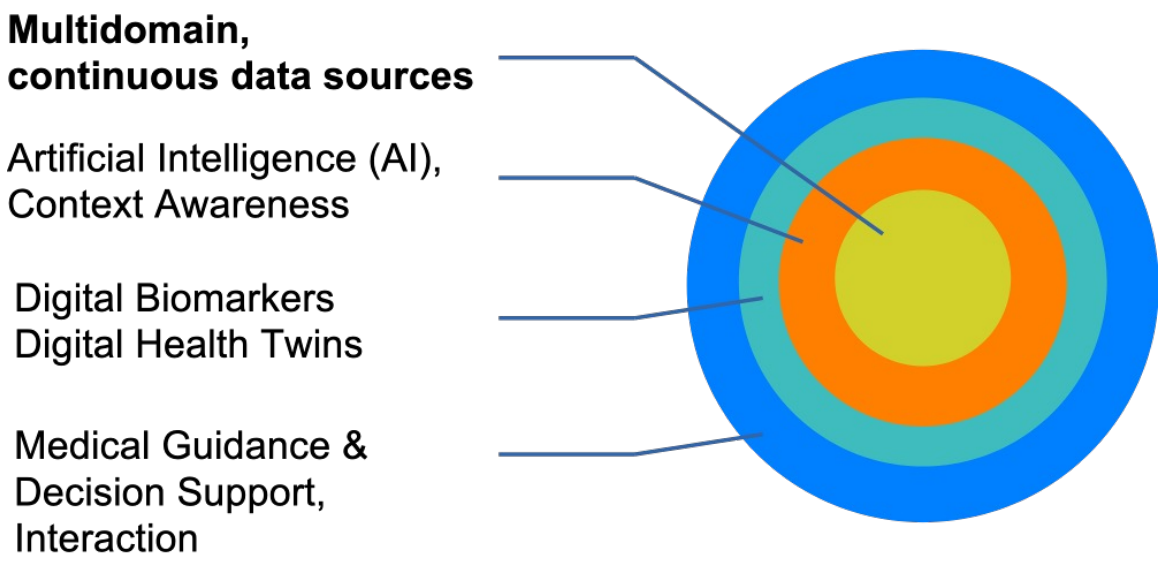
Clinical Decision-Support Command Suite

Patient Care at individual Departments, Patient Ward, ORs and ICU

Amato et al. “Central command suite: futureproofing next-generation surgical environments to embrace the digital operating room“, 2025.
Cypko et al. “Digital Health for Space: Towards Prevention, Training, Empowerment, and Autonomy“, 2025.

Agenda

1. Types of
   a) medical,
   b) ambient and
   c) mobile data
2. Balancing Data Collection and Value
3. Machanisms to increase data security and privacy



Understanding you.

1. What is health data?

2. Did you work with health data before?

3. What is your expectation to learn today?

4. What is your expectation to learn in the Digital Health lecture?

5. Are you looking for Master Projects or Thesis?

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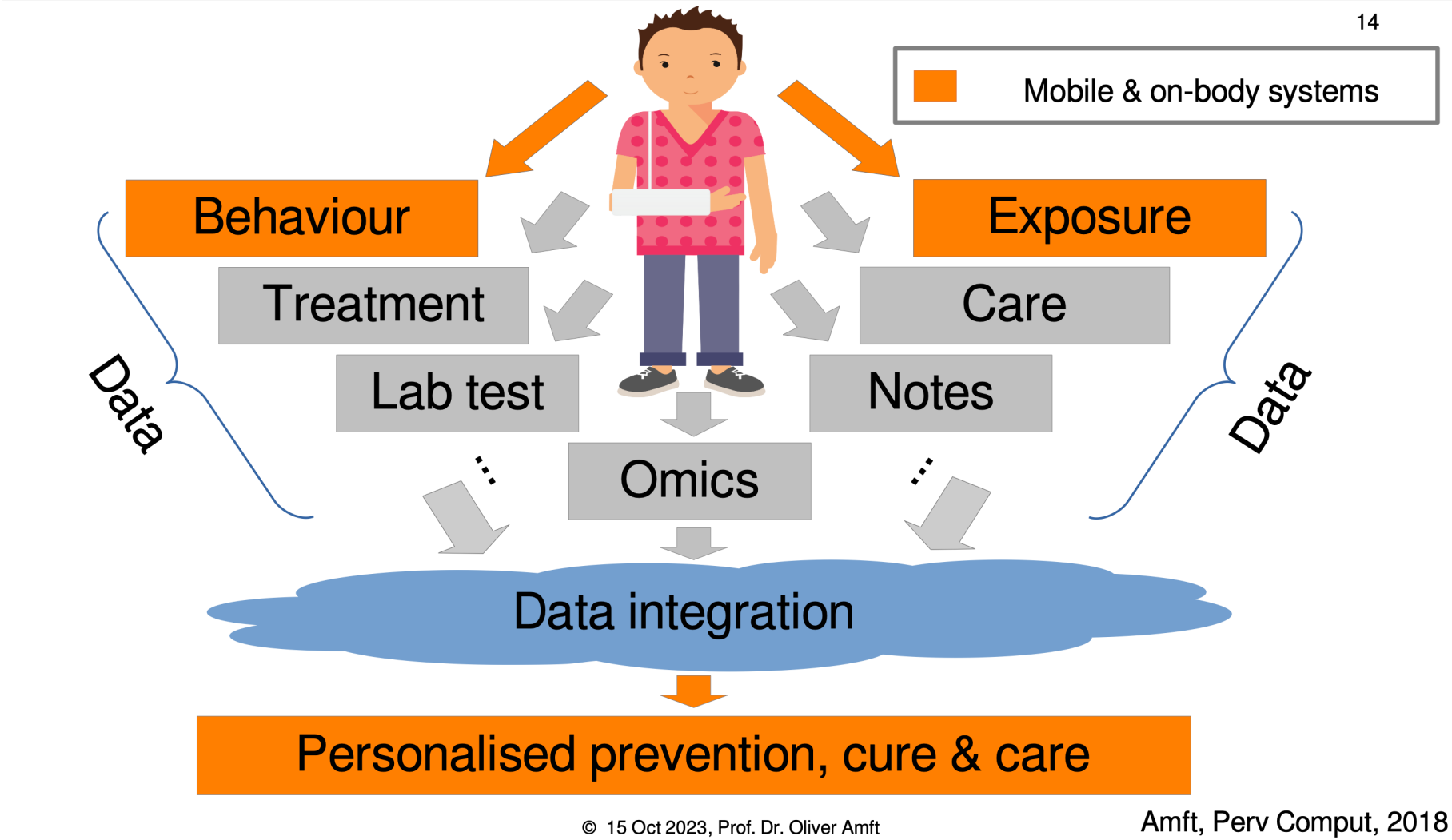
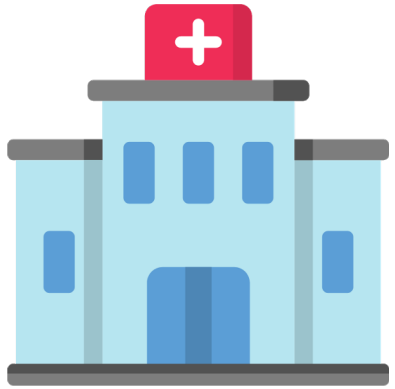
Reminder


14
Mobile & on-body systems
Behaviour
Exposure
Treatment
Care
Lab test
Notes
Omics
Data
Data
Data integration
Personalised prevention, cure & care
© 15 Oct 2023, Prof. Dr. Oliver Amft
Amft, Perv Comput, 2018

Types of medical, ambient and mobile data

Where is health data collected?

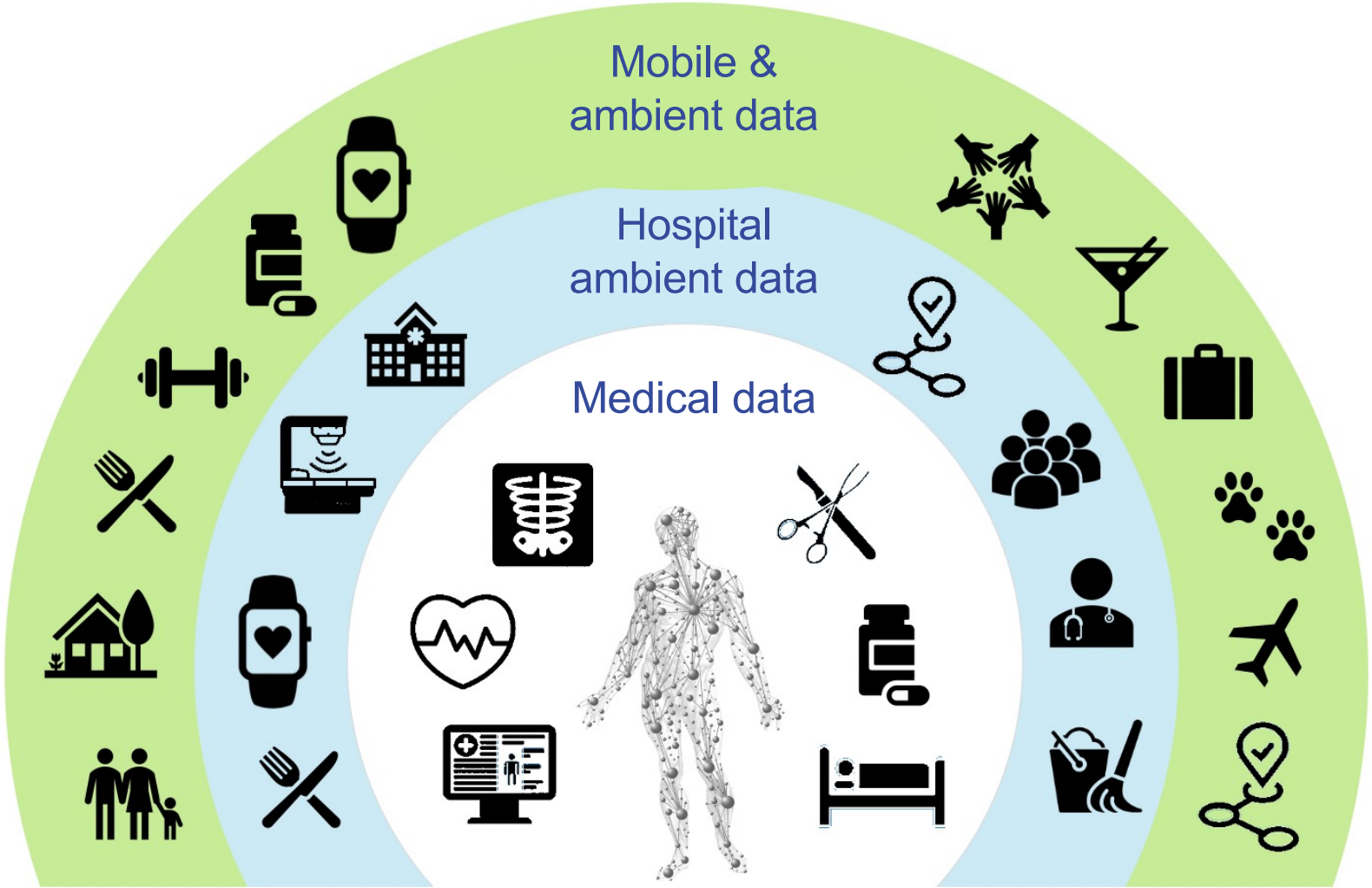
Medical Data



Mobile Data

Ambient Data

Scope of health and ambient data



Medical data

Photo by Owen Beard on Unsplash

Medical data landscape

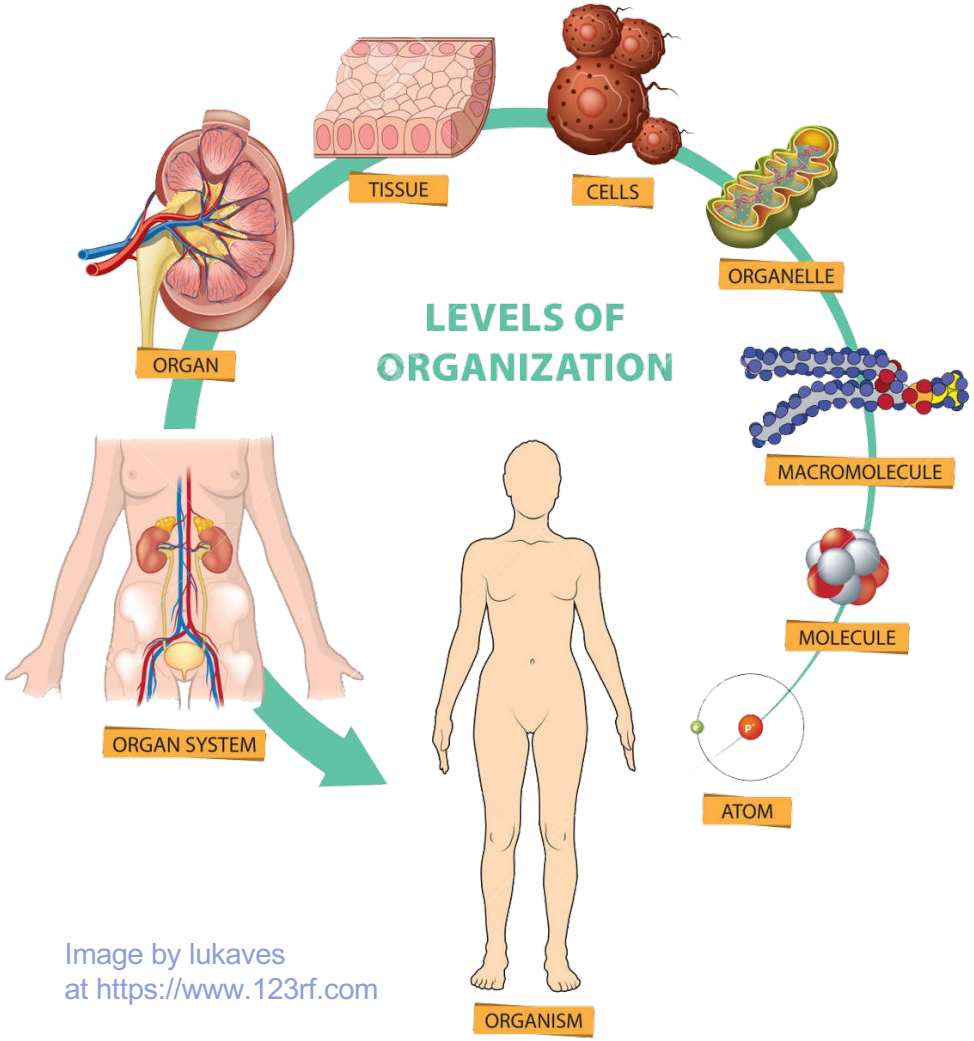
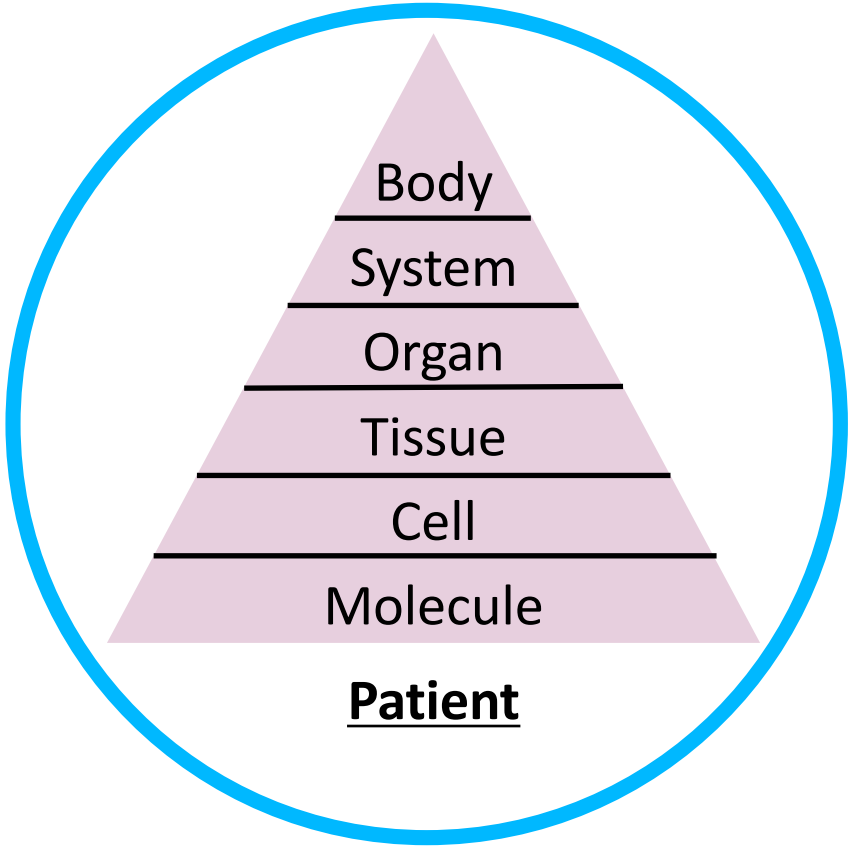


Image by lukaves
at https://www.123rf.com



Medical data landscape - Omics

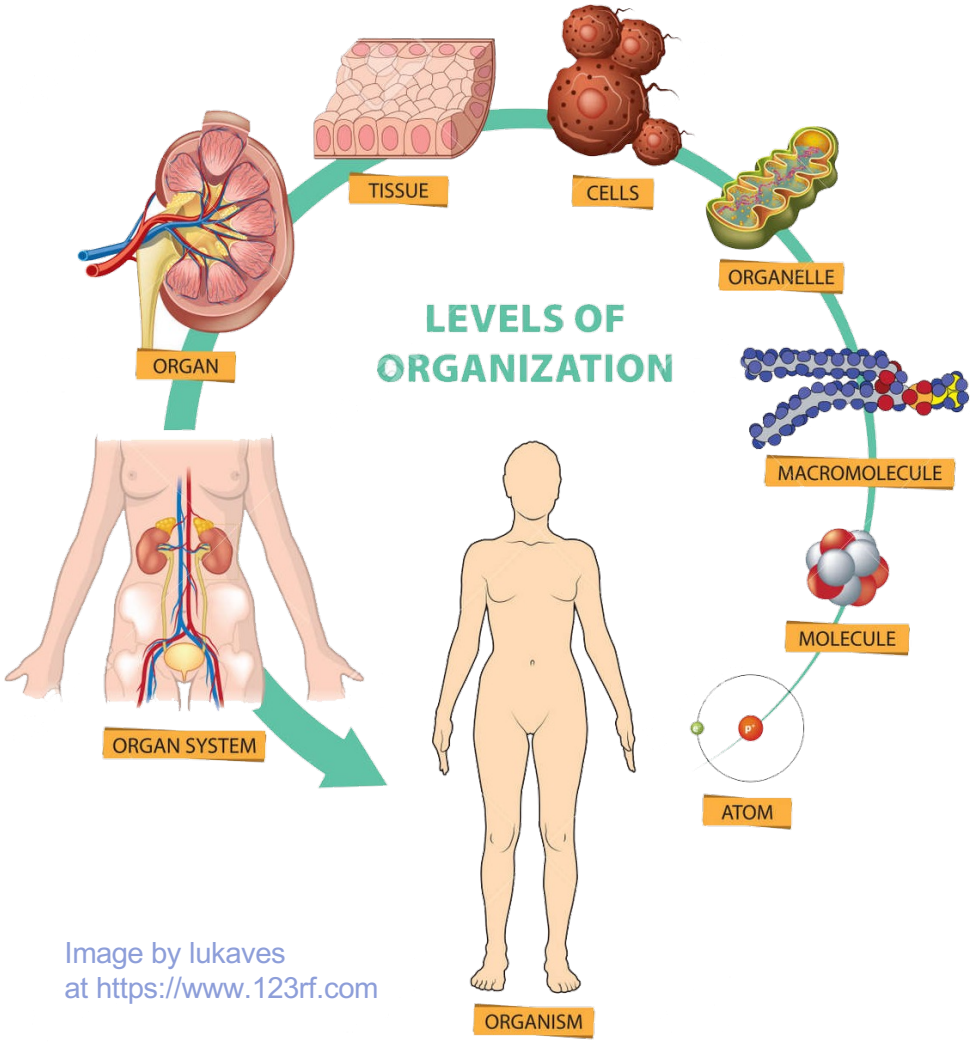


Image by lukaves
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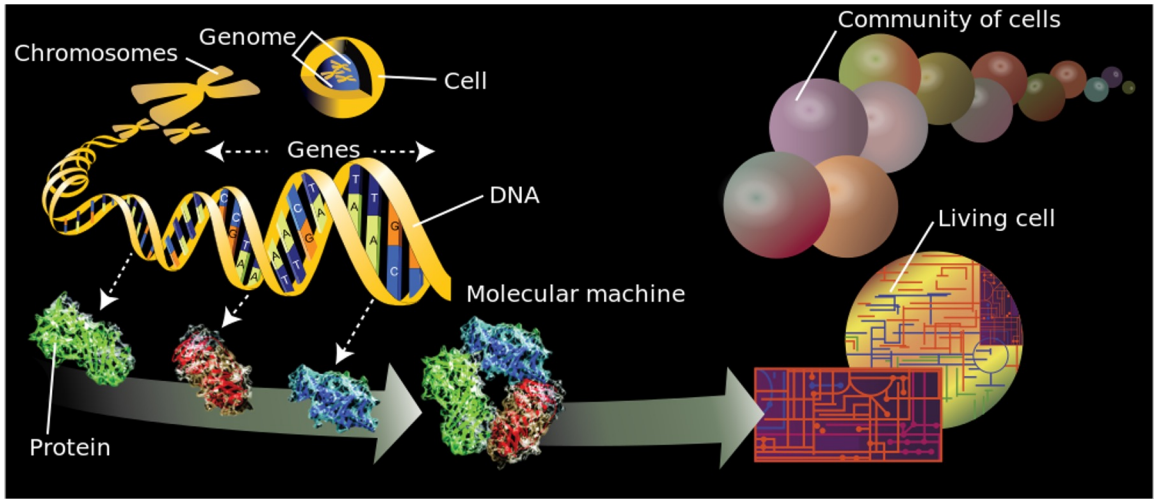


- **Genomics:** Study of the genomes of organisms, e.g. function.
- **Proteomics:** Large-scale study of proteins, particularly their structures and functions.
- **Metabolomics:** Scientific study of chemical processes involving metabolites.
- **Exposomic:** Study of the totality of environmental exposures throughout the lifetime and their impact on health

Wikipedia, Yale University

Medical data landscape – Examples of organ systems

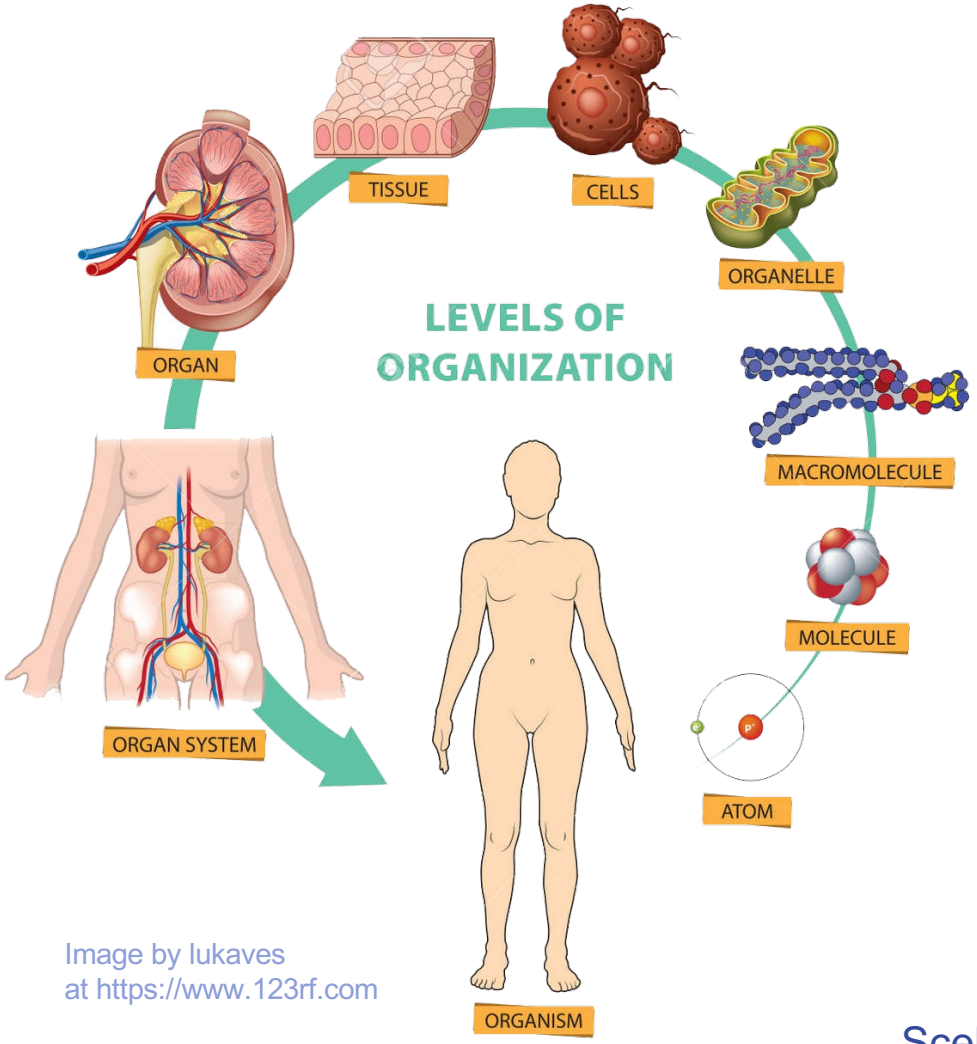
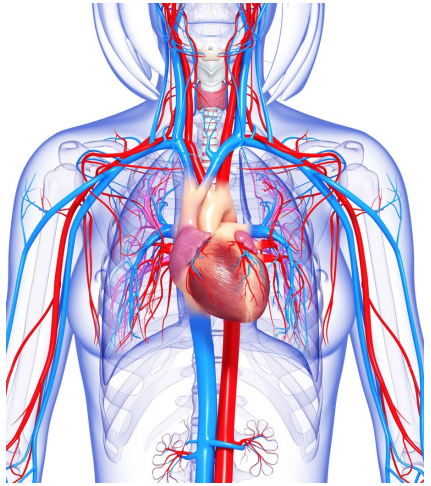
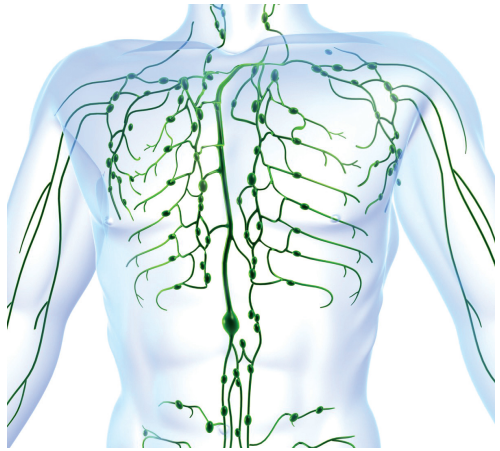


Image by lukaves
at https://www.123rf.com



Cardiovescular system



LIymphatic system

Sceloton system

Nervouse system

Digestive system

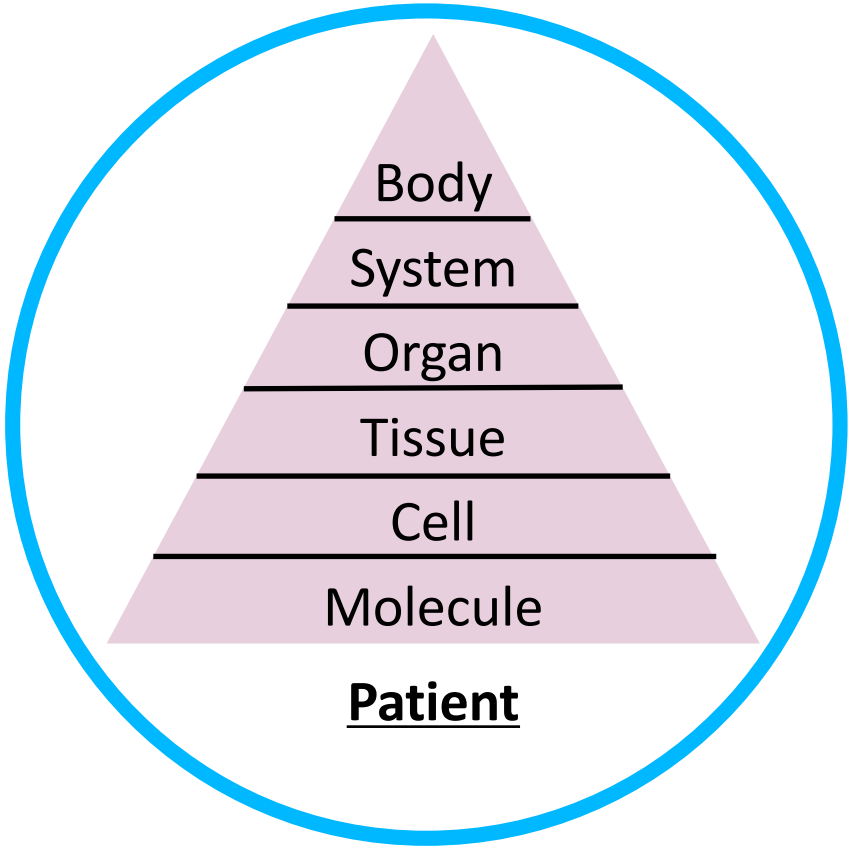
Medical data landscape

Anthropo-metrics:
Age, sex, height, weight

Exam notes:
Allergies
Diagnoses
Family history
Observations

Healthcare data:
Entry, discharge
Duration
Frequency

Treatment:
Vaccinations
Prescriptions
Hospitalisations
Surgeries



Behaviour/ in situ data:
Physical activity
Diet / nutrition
Sleep, Smoking
Medication
Param. monitoring

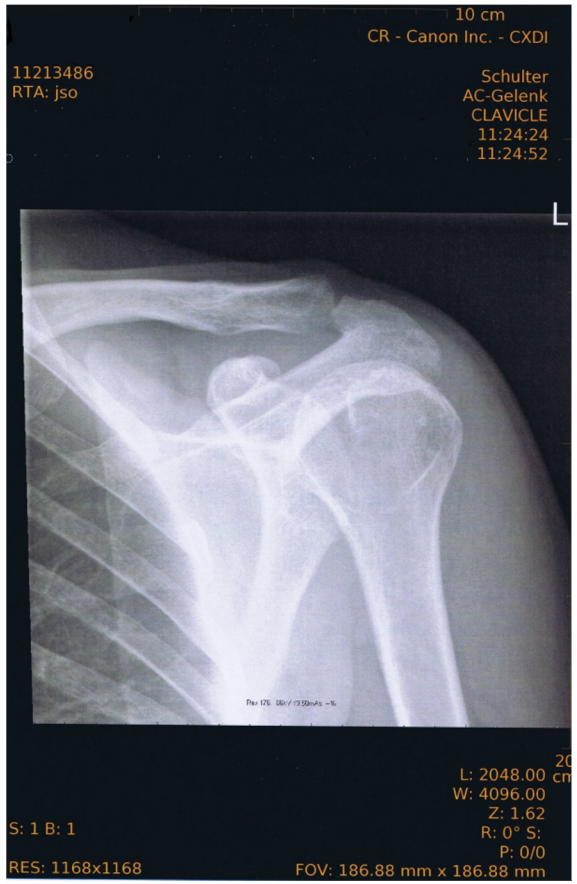
Lab data:
ICU, e.g. ECG
Sleep lab, Polysomnography

Lab test:
Blood test
Imaging

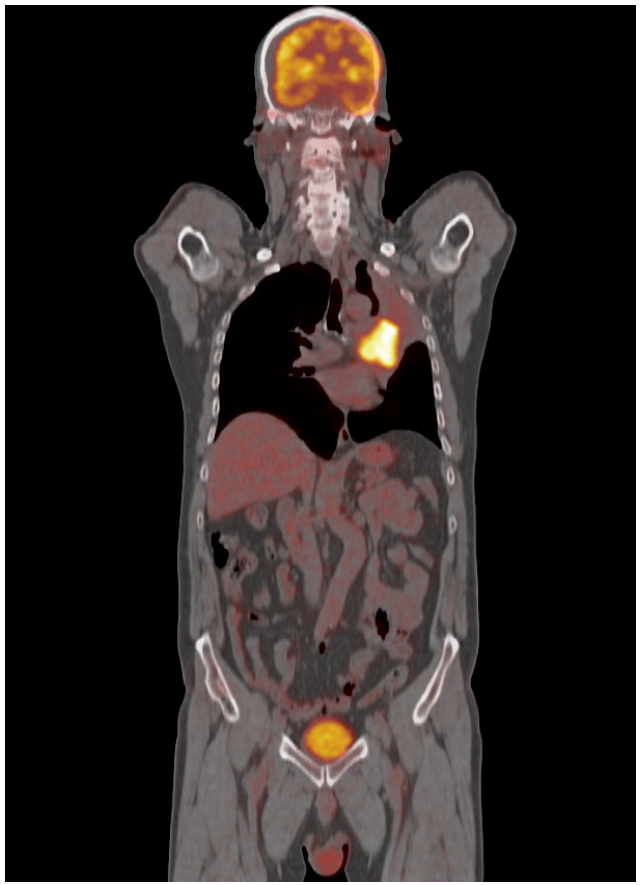
Omics:
Genomics
Proteomics

Medical data collection – Examples of medical devices

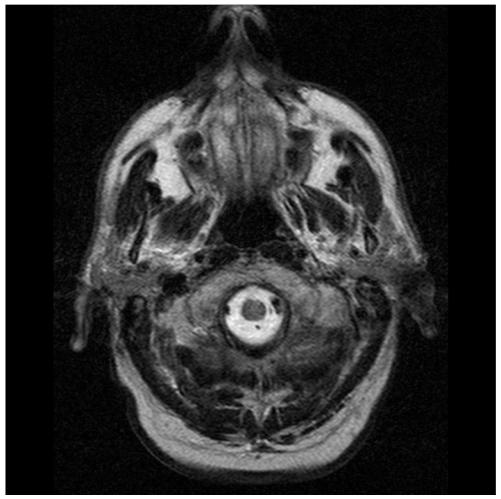
Medical (raw) data collection – Examples of Imaging

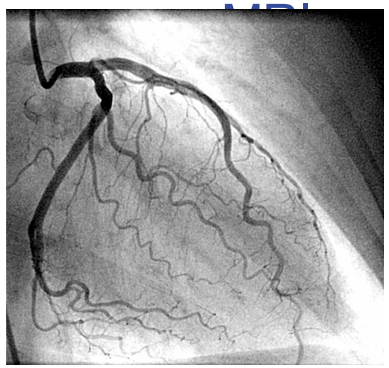


X ray

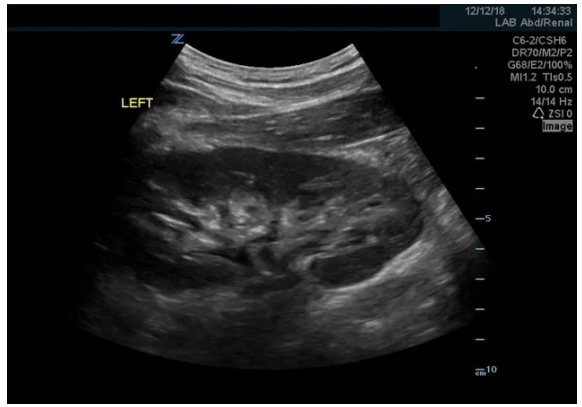


PET-CT

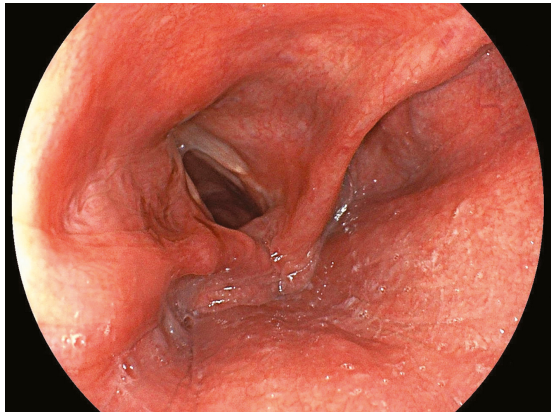




Angiography



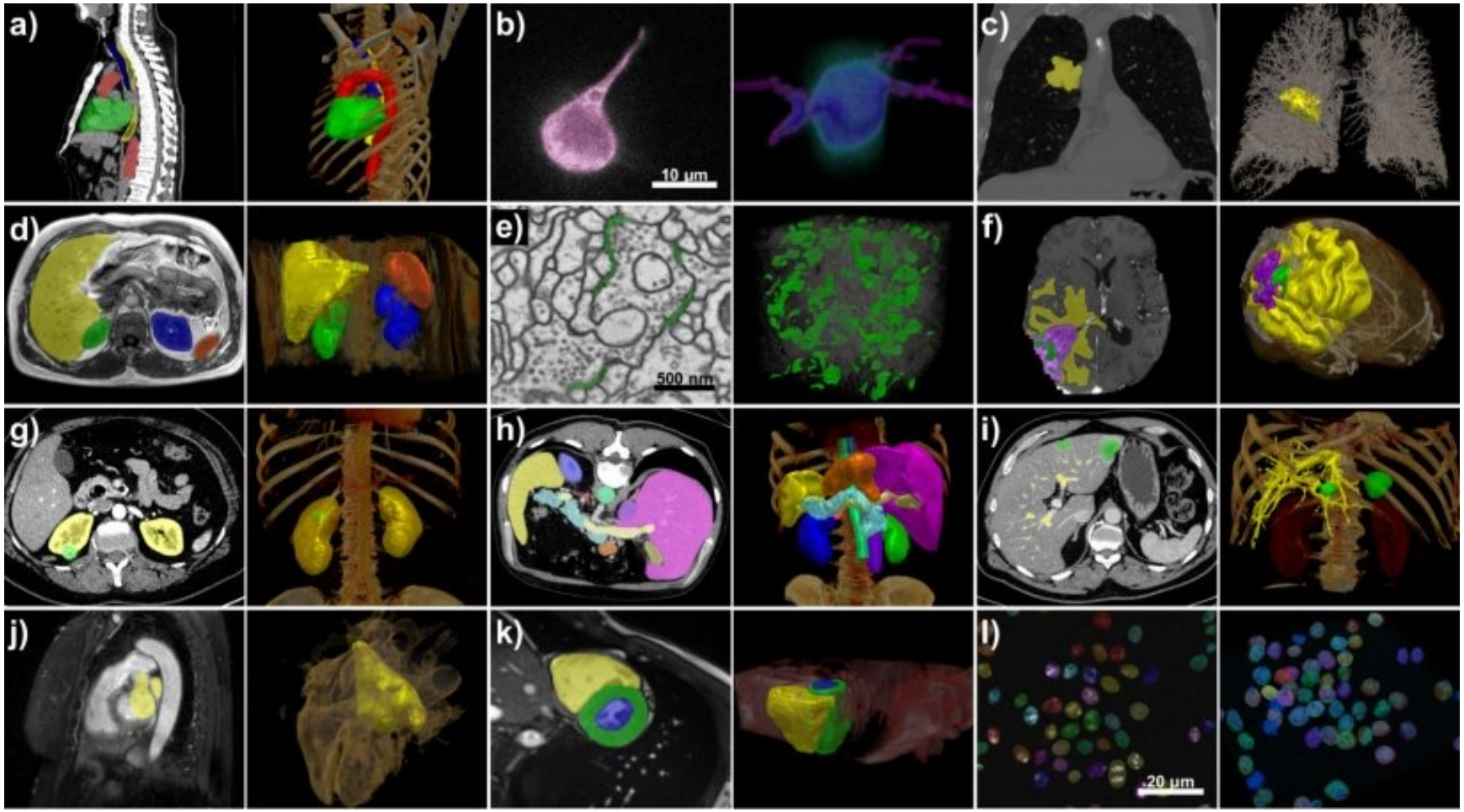
Ultrasound



Endoscopy

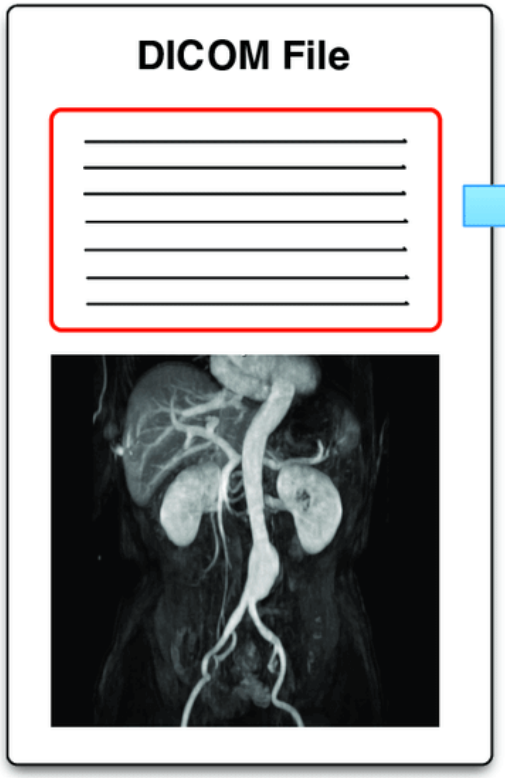
Examplary open access images from Google search.

Medical data collection – Examples of Imaging



Isensee et al. Nature Methods

Medical (raw) data collection – File formats



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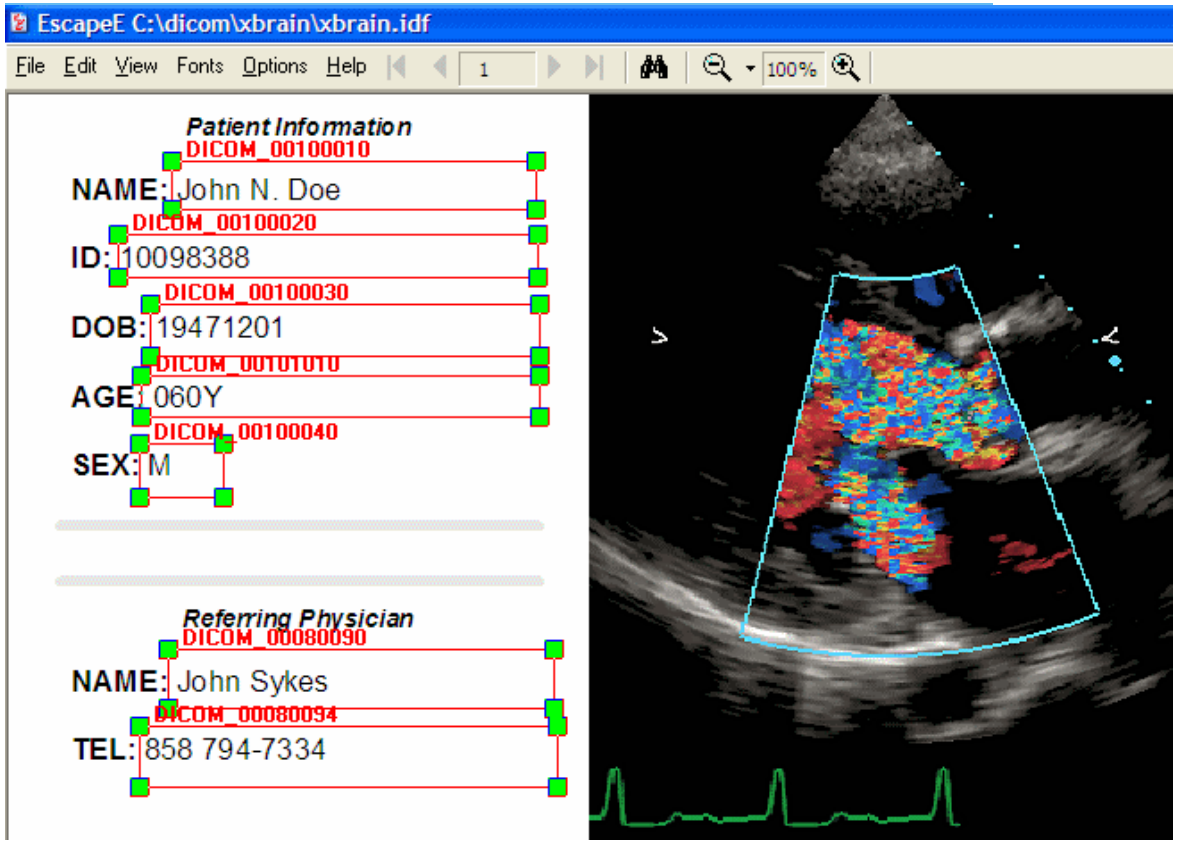
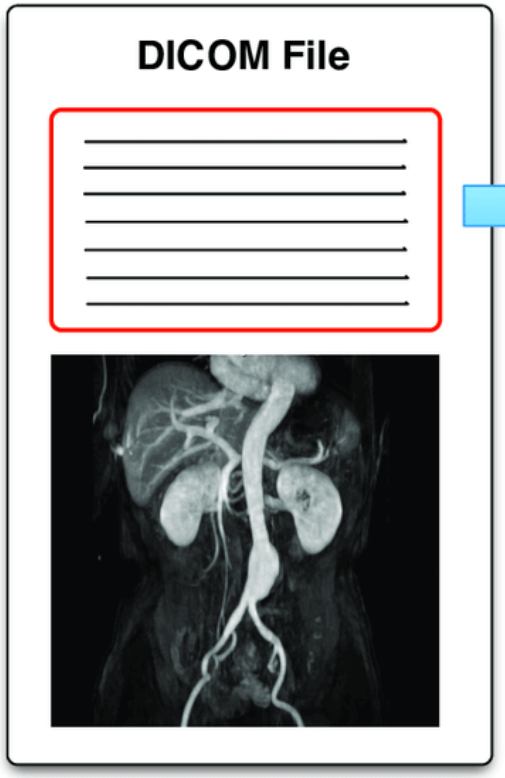
DICOM
Digital Imaging and Communications in Medicine

DICOM - Meta Data



Medical (raw) data collection – File formats



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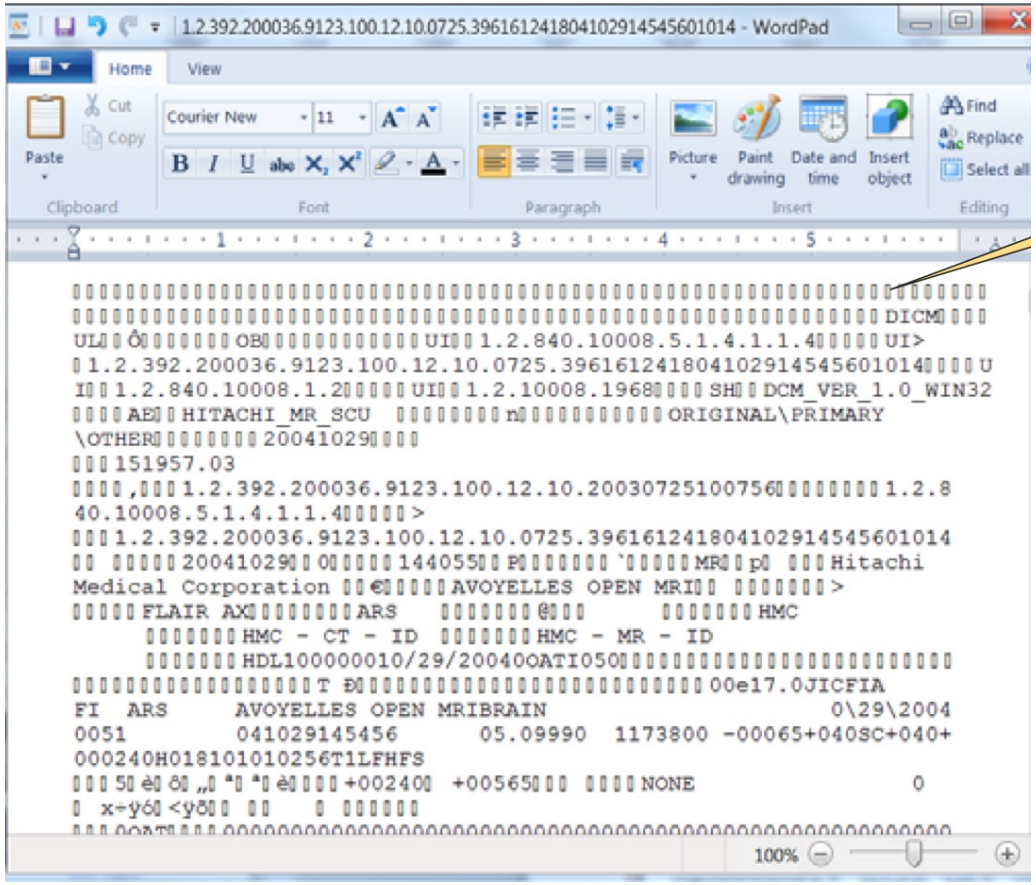
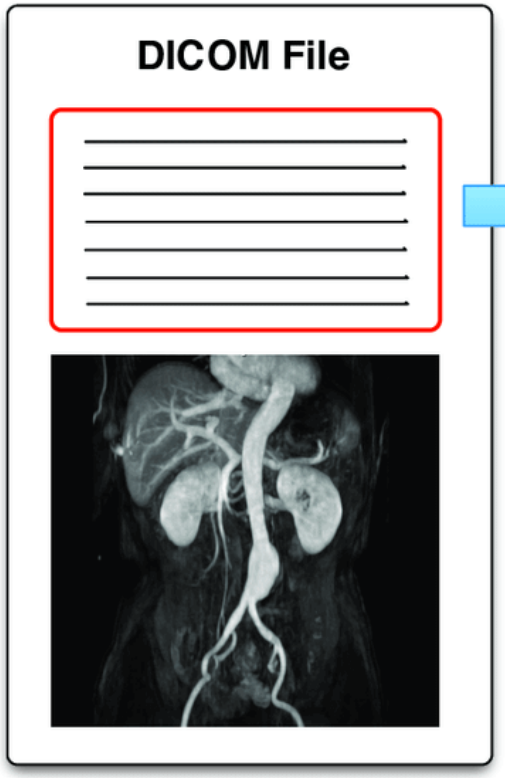
DICOM
Digital Imaging and Communications in Medicine

Old DICOM format



Medical (raw) data collection – File formats



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DICOM
Digital Imaging and Communications in Medicine

Newer XML-based DICOM formats

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Medical (raw) data collection – Example of an ECG

Image by www.ksmedizintechnik.de/

Medical (raw) data collection – Example of blood test

Von Labor LS-Medizin-Service Do 25 Feb 2010 15:54:39 CET Seite 1 of 4

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Per Fax an 0041618330714

0762330080

Laborgemeinschaft Albtal Otto-Hahn-Straße 18 76275 Ettlingen
Fax vom 25.02.2010/15:54 Form:.4 Tel: 07243-51603

Praxis 1099 Herrn Dr. med. Matthias Heisler
Schillerstr. 6 79618 Rheinfelden
Patient: 1099-6674/Amft Oliver P-Eing:15.01.2010 Abr.: LG
FAX-ENDBEFUND am 15.01.2010 Blatt 1 von 2

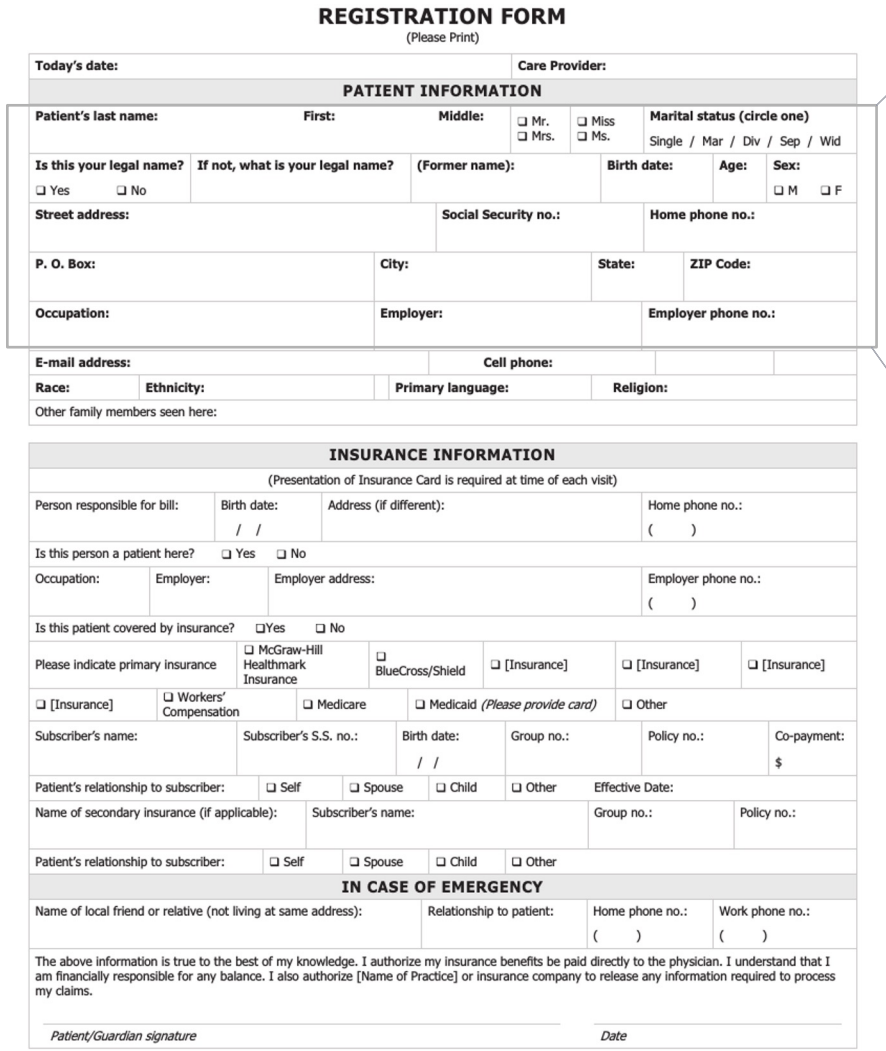
Ziff	Untersuchung	Messwert	Dimension	Referenzbereich	Grafik
	UROLOG.-PROFIL I ELEKTROLYTE				
KLINISCHE CHEMIE					
3555	Calcium im Serum	2.46	mMol/l 1)	2.00-2.65	[----*--]
3556	Chlorid im Serum	101	mMol/l	95-108	[---*---]
3620	Eisen im Serum	84	µg/dl	70-150	[-*-----]
3557	Kalium im Serum	5.25	mMol/l	3.60-5.50	[-----*-]
3558	Natrium im Serum	137	mMol/l	134-152	[-*-----]
3580H1	Phosphat im Serum	1.13	mMol/l	0.80-1.50	[---*---]
3592H1	Gamma-GT	21	U/l	< 60	[--*----]
3594H1	GOT	33	U/l	10-50	[---*---]
3595H1	GPT	27	U/l	10-50	[---*---]
3584H1	Harnstoff	29	mg/dl	10-45	[---*---]
3583H1	Harnsäure	6.8	mg/dl	2.6-6.8	[-------*]
3585H1	Kreatinin	1.05	mg/dl	0.70-1.30	[---*---]
3560	Glucose im Serum	80	mg/dl 2)	60-110	[--*----]
3561	HbAlc	5.3	% 3)	3.9-5.7	[-----*-]
GERINNUNG					
3605	PTT	32	sec	25-37	[---*---]
3607	Quick (Prozentwert)	97	%	70-130	[---*---]
	INR (Quick als INR)	1.01	INR	< 1.20	[-----*-]
3606	Thrombinzeit (TZ)	14	sec	10-17	[----*--]
HÄMATOLOGIE					
3550	KLEINES BLUTBILD mit THROMBOZYTEN Leukozyten	6.64	/nl	4.30-10.80	[--*----]

Image by Prof. Amft

KLINISCHE CHEMIE					
3555	Calcium im Serum	2.46	mMol/l 1)	2.00-2.65	[----*--]
3556	Chlorid im Serum	101	mMol/l	95-108	[---*---]
3620	Eisen im Serum	84	µg/dl	70-150	[-*-----]
3557	Kalium im Serum	5.25	mMol/l	3.60-5.50	[-----*-]
3558	Natrium im Serum	137	mMol/l	134-152	[-*-----]
3580H1	Phosphat im Serum	1.13	mMol/l	0.80-1.50	[---*---]

Note: The choice of unit of measurement, decimal point and number of digits after the decimal point may be different, depending on the laboratory.

Medical (raw) data collection – Example of patient questionnaire



REGISTRATION FORM

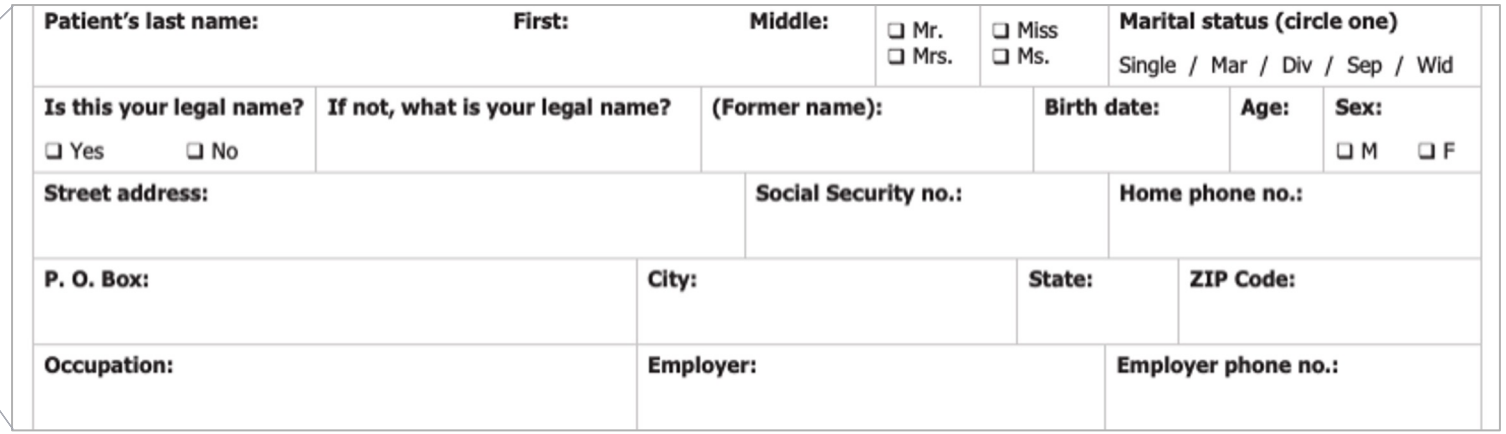
(Please Print)

Today's date:			**Care Provider:**			
PATIENT INFORMATION						
Patient's last name:	**First:**	**Middle:**	❑ Mr. ❑ Mrs.	❑ Miss ❑ Ms.	**Marital status (circle one)** Single / Mar / Div / Sep / Wid	
Is this your legal name? ❑ Yes ❑ No	**If not, what is your legal name?**	**(Former name):**	**Birth date:**	**Age:**	**Sex:** ❑ M ❑ F	
Street address:		**Social Security no.:**	**Home phone no.:**			
P. O. Box:	**City:**	**State:**	**ZIP Code:**			
Occupation:	**Employer:**	**Employer phone no.:**				
E-mail address:		**Cell phone:**				
Race:	**Ethnicity:**	**Primary language:**	**Religion:**			
Other family members seen here:						

INSURANCE INFORMATION					
(Presentation of Insurance Card is required at time of each visit)					
Person responsible for bill:	Birth date: / /	Address (if different):	Home phone no.: ( )		
Is this person a patient here? ❑ Yes ❑ No					
Occupation:	Employer:	Employer address:	Employer phone no.: ( )		
Is this patient covered by insurance? ❑Yes ❑ No					
Please indicate primary insurance	❑ McGraw-Hill Healthmark Insurance	❑ BlueCross/Shield	❑ [Insurance]	❑ [Insurance]	❑ [Insurance]
❑ [Insurance]	❑ Workers' Compensation	❑ Medicare	❑ Medicaid *(Please provide card)*	❑ Other	
Subscriber's name:	Subscriber's S.S. no.:	Birth date: / /	Group no.:	Policy no.:	Co-payment: $
Patient's relationship to subscriber:	❑ Self	❑ Spouse	❑ Child	❑ Other	Effective Date:
Name of secondary insurance (if applicable):	Subscriber's name:		Group no.:	Policy no.:	
Patient's relationship to subscriber:	❑ Self	❑ Spouse	❑ Child	❑ Other	
IN CASE OF EMERGENCY					
Name of local friend or relative (not living at same address):	Relationship to patient:	Home phone no.: ( )	Work phone no.: ( )		

The above information is true to the best of my knowledge. I authorize my insurance benefits be paid directly to the physician. I understand that I am financially responsible for any balance. I also authorize [Name of Practice] or insurance company to release any information required to process my claims.

Patient/Guardian signature *Date*



Patient's last name:	**First:**	**Middle:**	❑ Mr. ❑ Mrs.	❑ Miss ❑ Ms.	**Marital status (circle one)** Single / Mar / Div / Sep / Wid	
Is this your legal name? ❑ Yes ❑ No	**If not, what is your legal name?**	**(Former name):**	**Birth date:**	**Age:**	**Sex:** ❑ M ❑ F	
Street address:		**Social Security no.:**	**Home phone no.:**			
P. O. Box:	**City:**	**State:**	**ZIP Code:**			
Occupation:	**Employer:**	**Employer phone no.:**				

- Open-ended Text (Free-text)
- Discrete Values (Categorical Data)
- Numerical Values
- Binary data
- Likert Scale
- Multiple Choice
- Date & Time

Medical (raw) data collection – FHIR File Format

```
<Patient xmlns="http://hl7.org/fhir">
  <extension url="http://www.goodhealth.org/consent#trials">
    <valueCode value="renal"/>
  </extension>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Henry Levin the 7th</p>
      <p>MRN: 123456</p>
    </div>
  </text>
  <identifier>
    <use value="usual"/>
    <label value="MRN"/>
    <system value="http://www.goodhealth.org/identifiers/mrn"/>
    <value value="123456"/>
  </identifier>
  <name>
    <family value="Levin"/>
    <given value="Henry"/>
    <suffix value="The 7th"/>
  </name>
  <gender>
    <text value="Male"/>
  </gender>
  <birthDate value="1932-09-24"/>
  <managingOrganization>
    <reference value="Organization/2"/>
    <display value="Good Health Clinic"/>
  </managingOrganization>
  <active value="true"/>
</Patient>
```

Extension with URL to definition

Human Readable Summary

Standard Data:
- MRN
- Name
- Gender
- Birth Date
- Provider

HL7® FHIR®

https://build.fhir.org/resourcelist.html

http://connect.mcgraw-hill.com

Medical (raw) data collection – Example of a surgical sketch

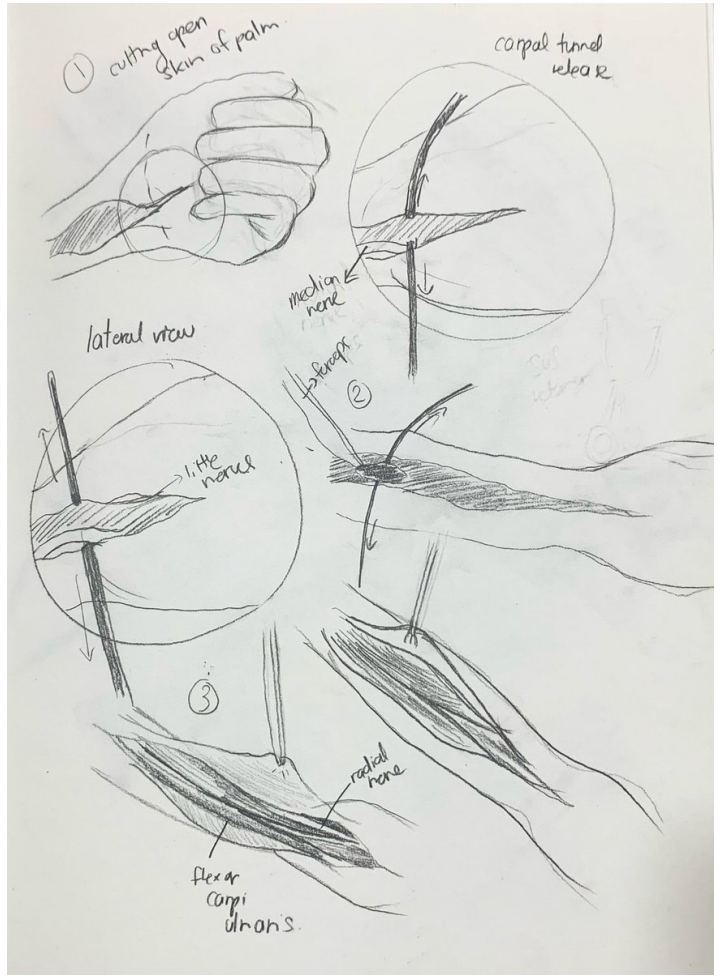
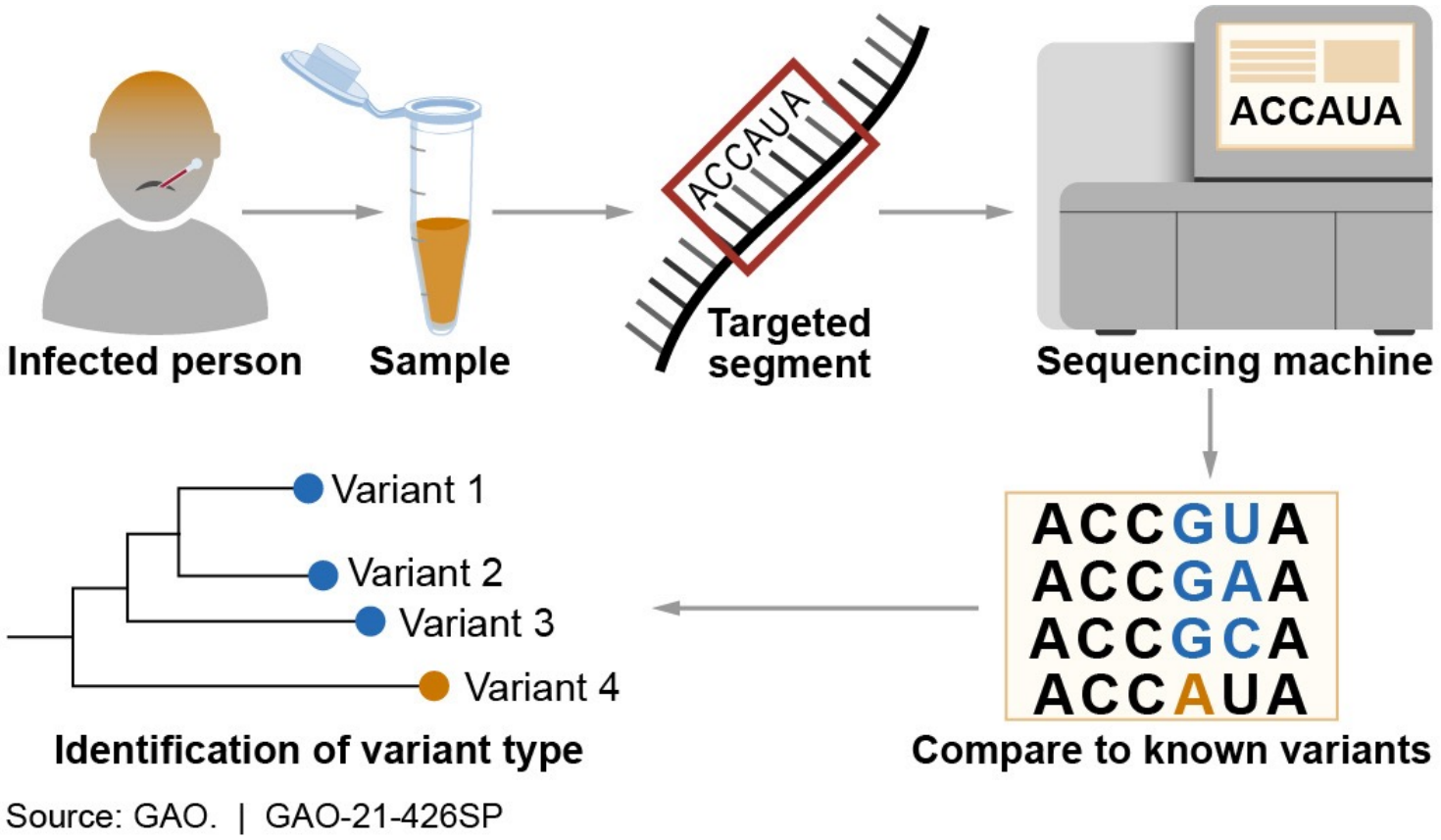


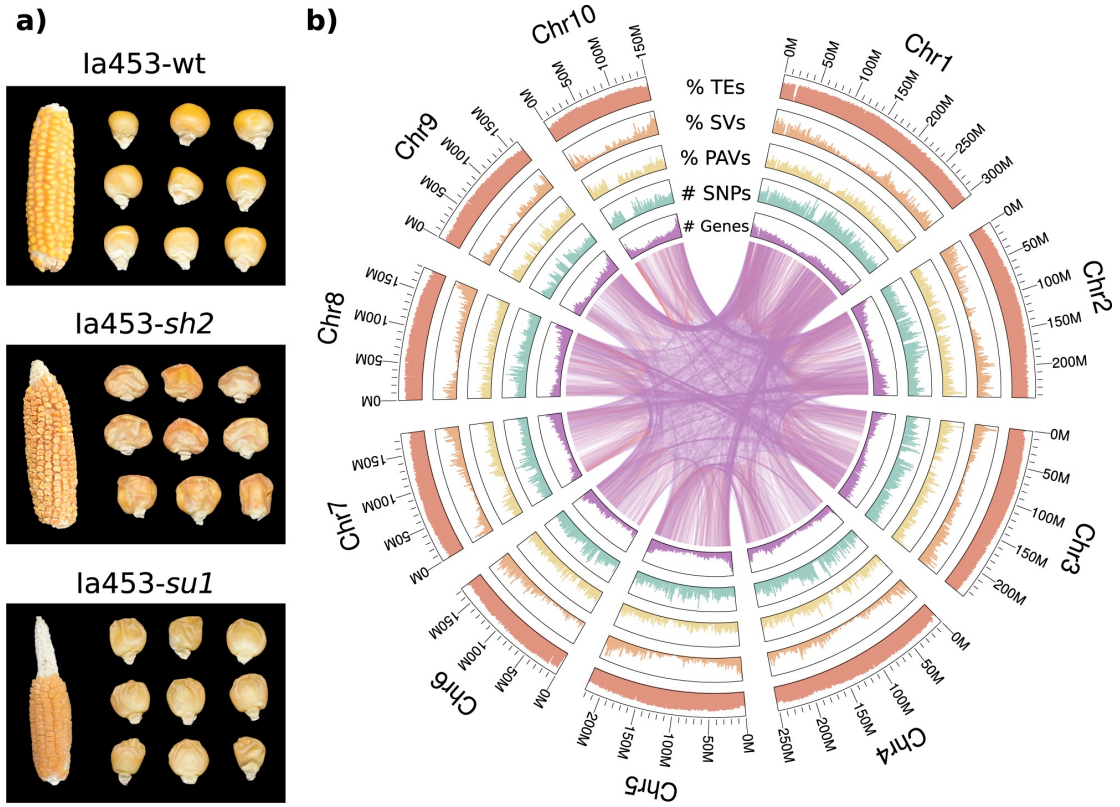
Image by sana @sfkhanvisuals on X

Medical (raw) data collection – Example of Genomic Sequencing

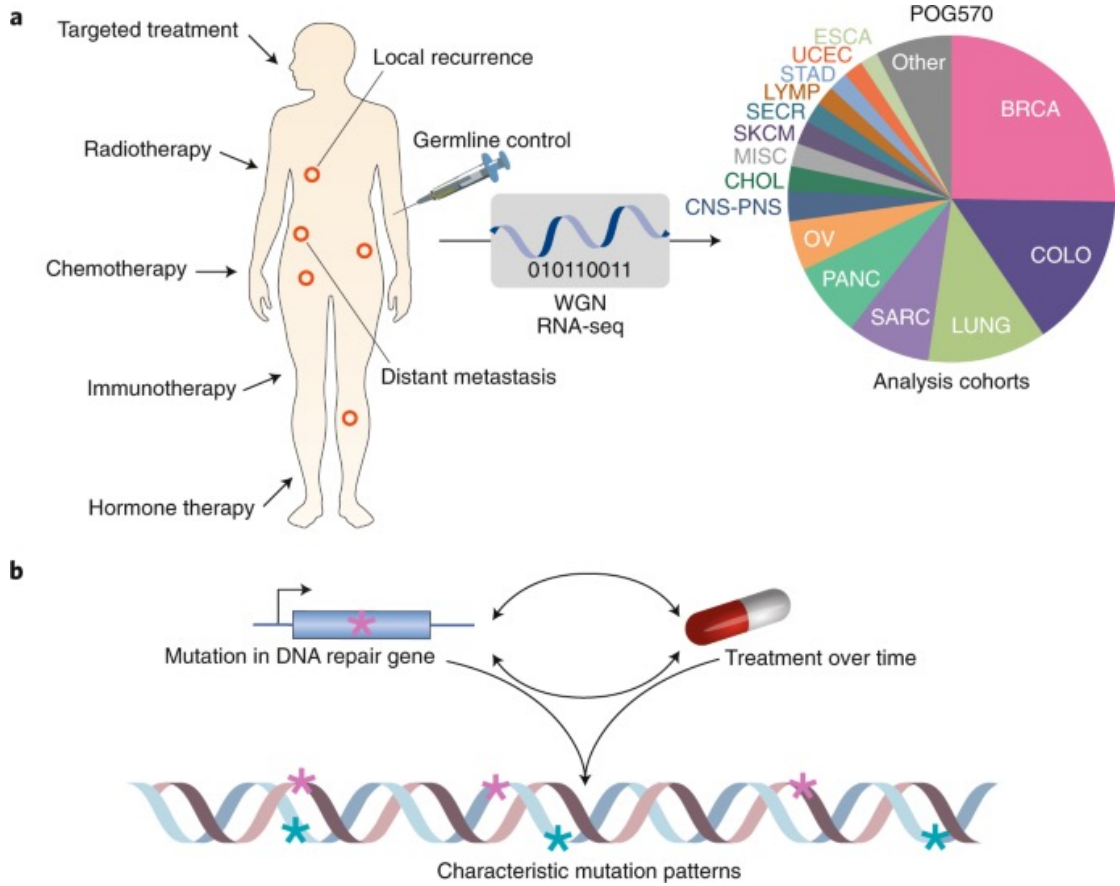


https://www.gao.gov/products/gao-21-426sp

Medical (raw) data collection – Example of Genomic Sequencing

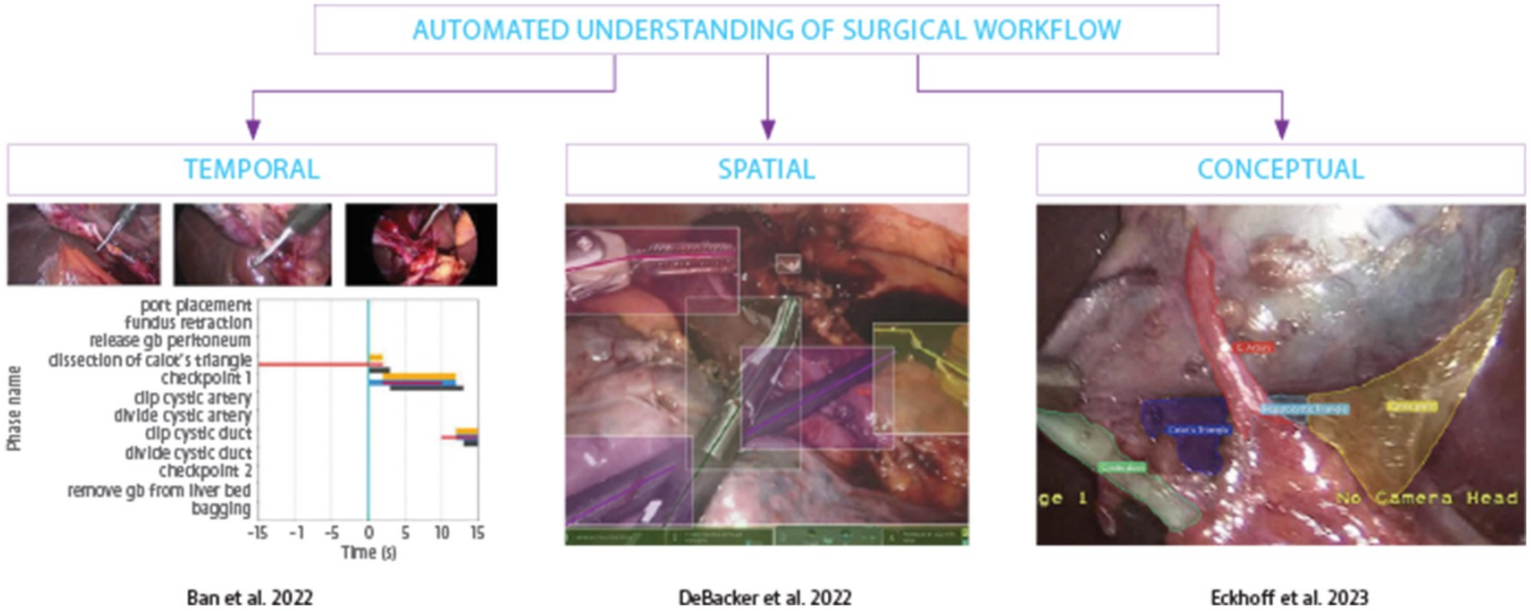


Hu, Ying, et al. "Genome assembly and population genomic analysis provide insights into the evolution of modern sweet corn." *Nature Communications* 12.1 (2021): 1227.



Rheinbay, Esther. "The genomic landscape of advanced cancer." *Nature Cancer* 1.4 (2020): 372-373.

Workflow data



Ban et al. 2022 DeBacker et al. 2022 Eckhoff et al. 2023

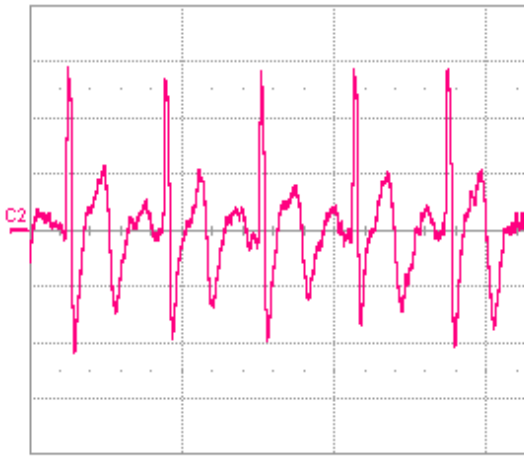
J. McCartney, infographic on surgical workflows. Bulletin 2023.

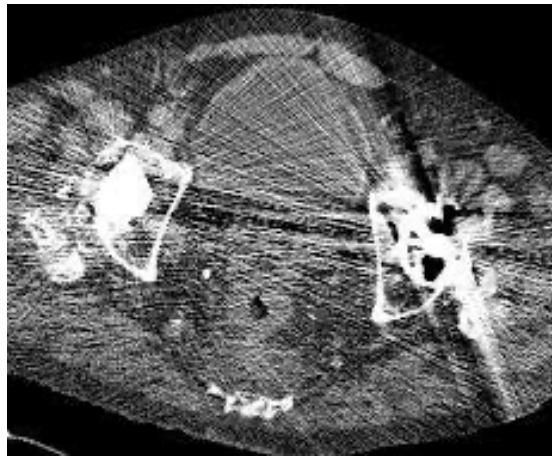
Workflow prediction to provide medical care at the right time.

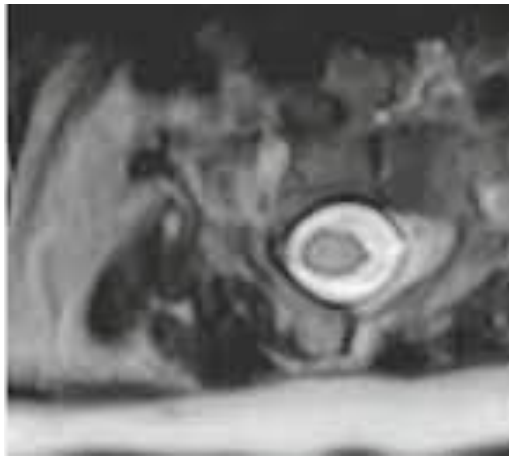
Medical (raw) data collection – Mistakes and Artifacts

- Questionnairs and messurements can be inaccurate
- Images can have noises artifacts
- Sensor data can have noises

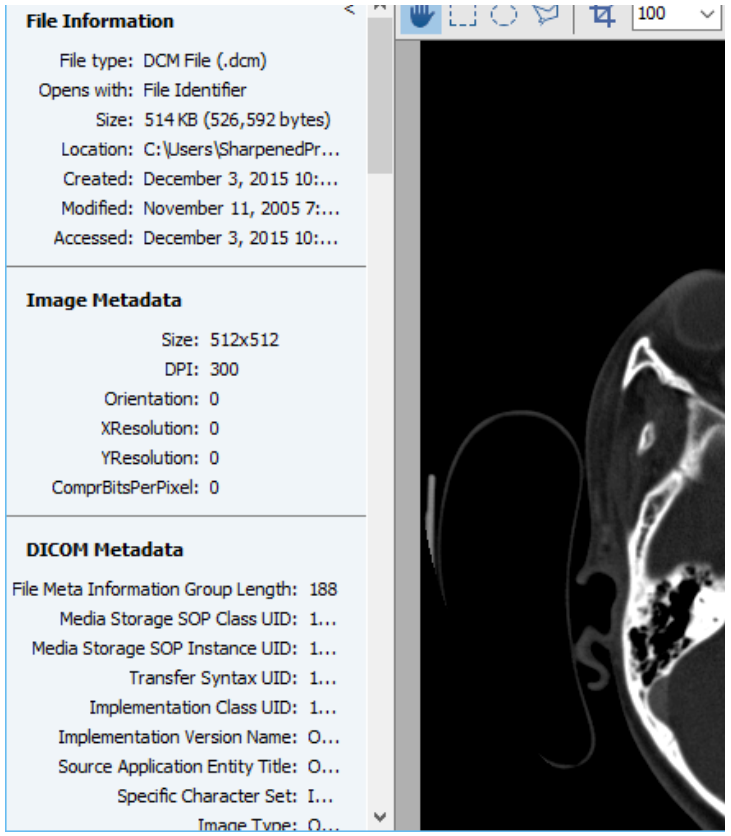
Note: Measurements of the same modality but different devices may be difficult to compare.







DICOM - Meta Data



Activation!

6. What inner body systems do you know?

7. What medical devices do you know?

8. What (clinical) data types do you know?

9. What are data standards and what are they for?

10. What are artifacts in data?

pingo.coactum.de/815896

Medical data summary

- Medical data point (/ datum): any single observation of a patient, defined by four elements:
  - Patient in question,
  - Parameter being observed,
  - Value of the parameter,
  - Time of observation.

- Note: The datum may have an uncertainty associated to it; sampling strategies apply.

Medical data summary

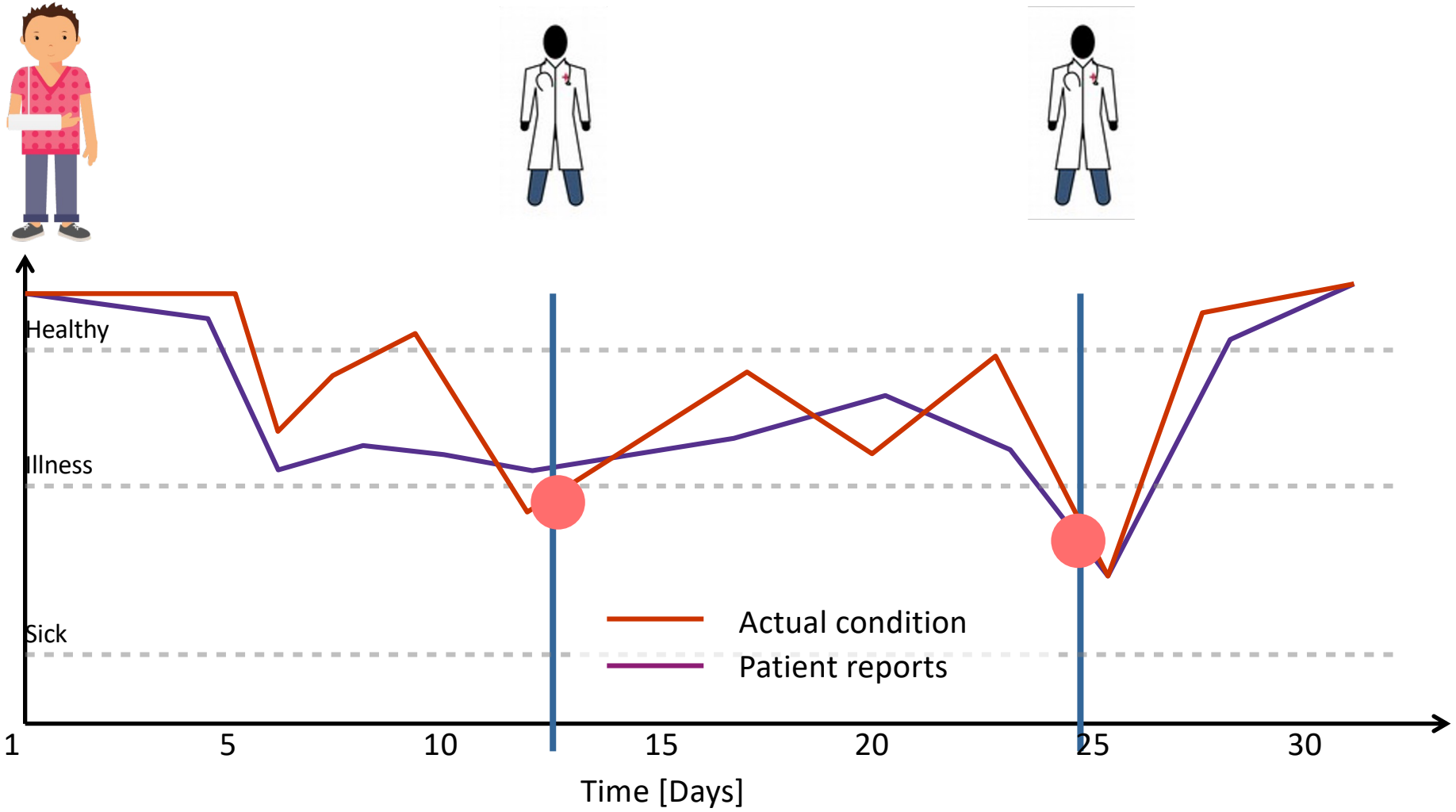
- Medical data: collection of observations.
- Types of medical data:
  - Narrative, text, coding: diagnosis, conditions, etc.
  - Numerical measurements: lab tests, e.g. blood test, vital signs.
  - Recorded signals: timeseries, e.g. ECG.
  - Images: e.g. radiology images, sketches.

Mobile data

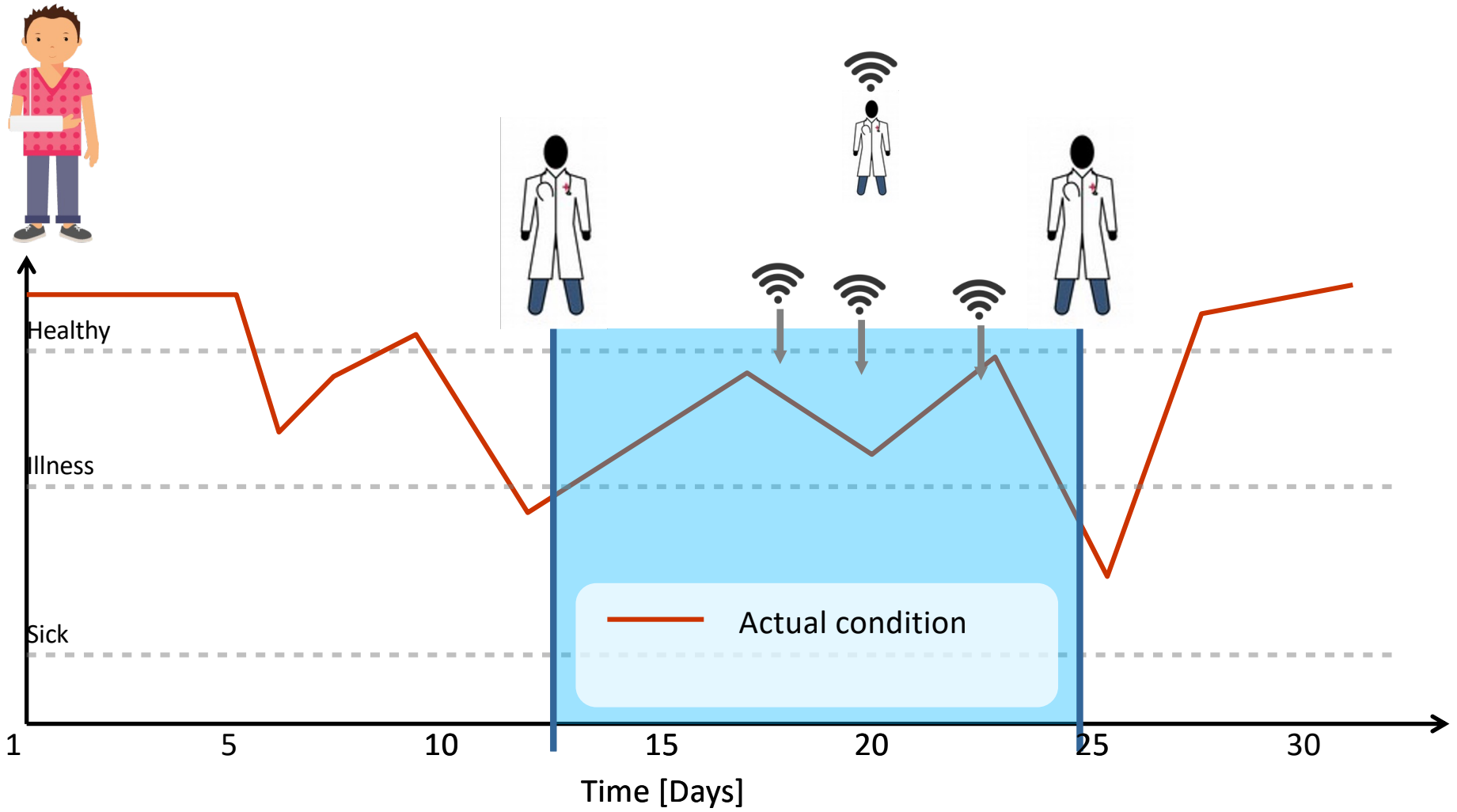
Motivation for mobile data

Image by Cypko, ICCAS

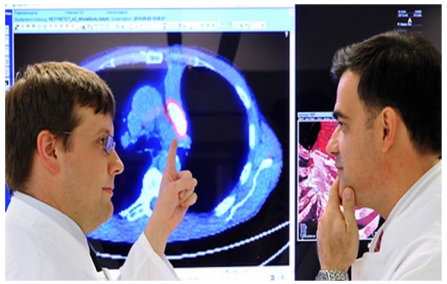
A paradigm shift



A paradigm shift

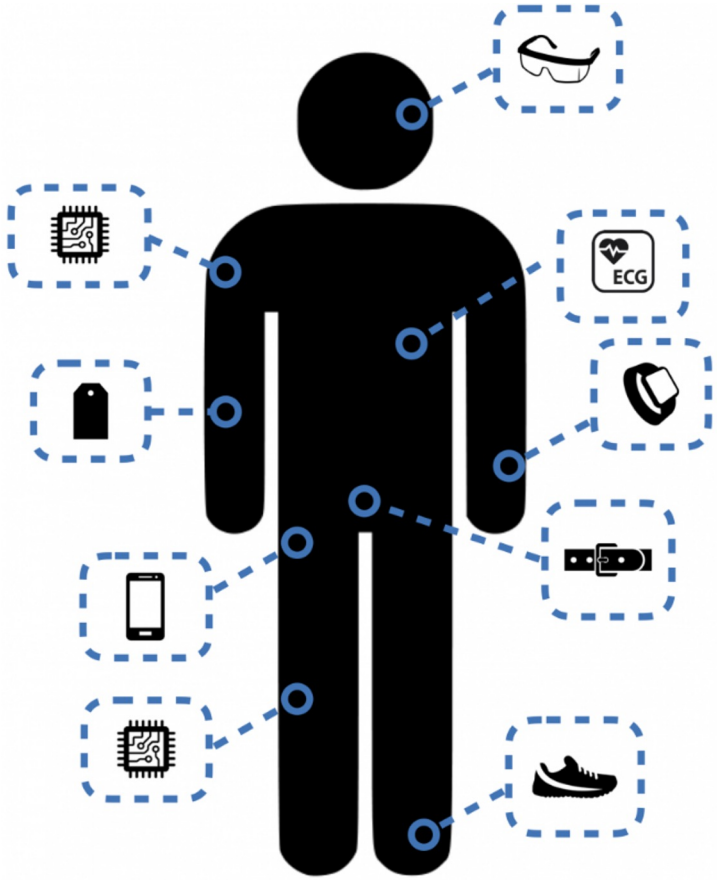


Mind gap – Missing mobile and ambient data



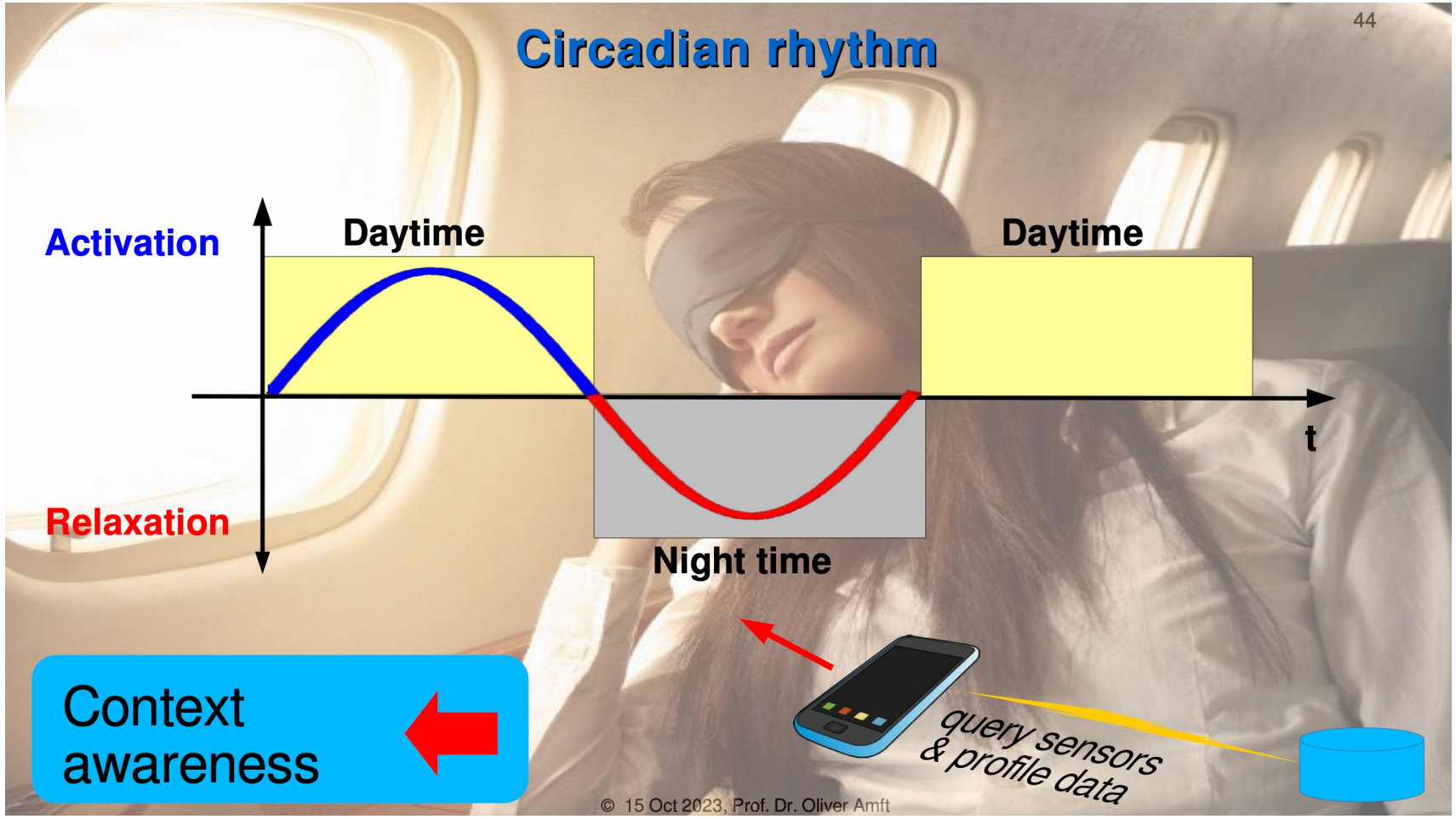
Source: new technology for old people. The Economist.

Mobile data



Images of Google devices

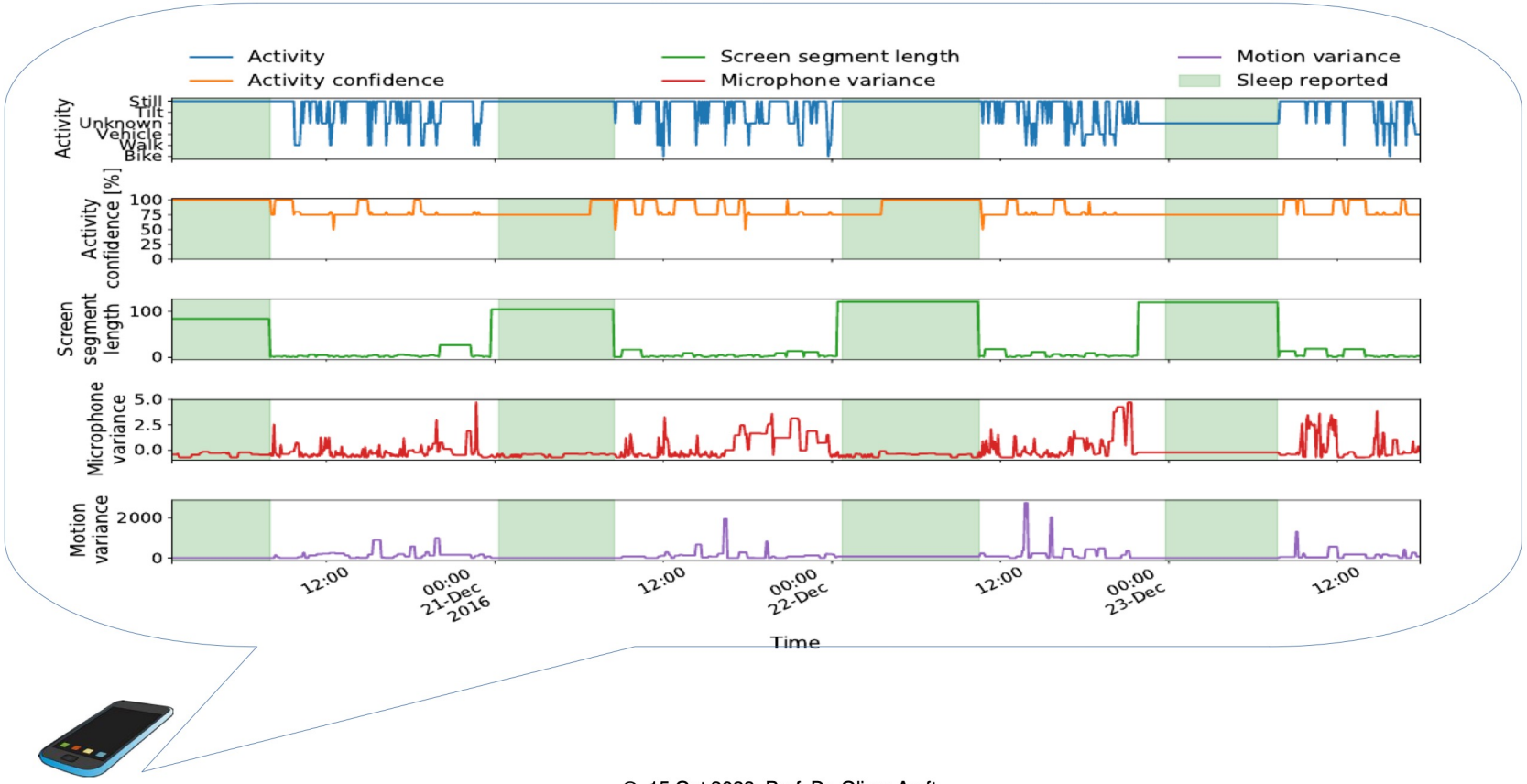
Mobile data – Sensors example



Mobile data – Sensors example

45

Context sampling on smartphone



© 15 Oct 2023, Prof. Dr. Oliver Amft

Mobile data - Sensors

Smart ring (Oura)

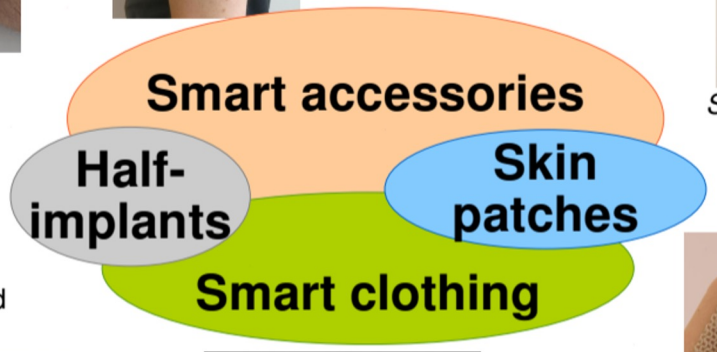
Smart phoes

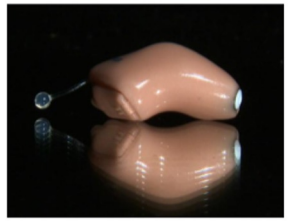
Smart watch (~2011)

QBIC: Q-Belt Integrated Computer (2004)

SimpleSkin: Textile sensor building blocks (2017)

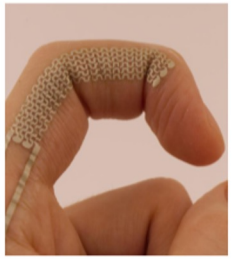
BodyMedia arm band (~2010)





Widex in-ear hearing aid (2015)

Smart eyeglasses (2015-17)



Printed interactive skin electronics (2017, Courtesy Prof. Steimle).

Amft & v Learhoven, IEEE Perv Comput 16(4), '17

Blood sugar

➔ Possiblity for continues measurements → Big data

Data Architecture

[Wikipedia]

Sensors

Signal processing

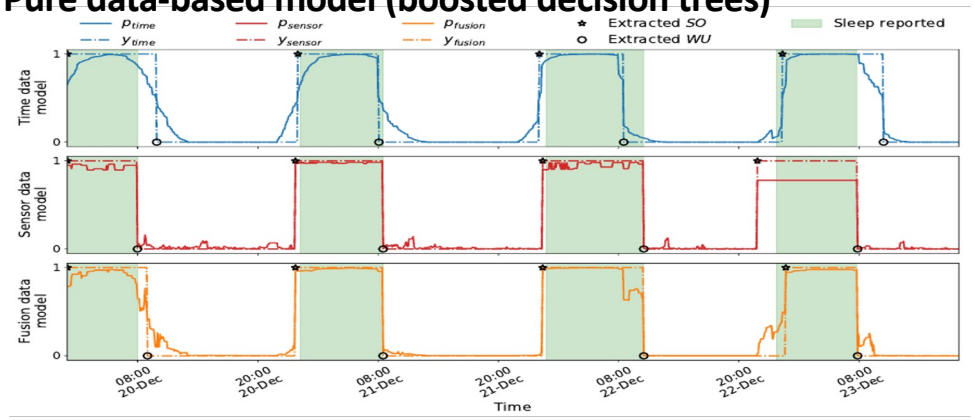
Health
HUBS

Cloud Support

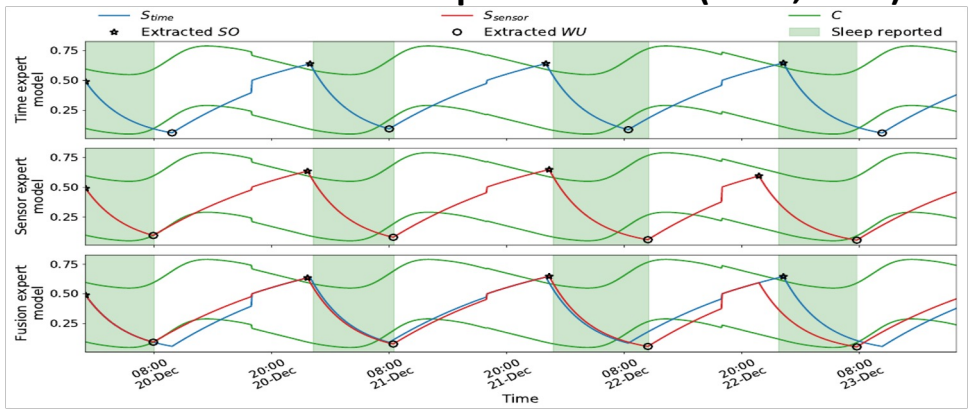
Visualization
UI

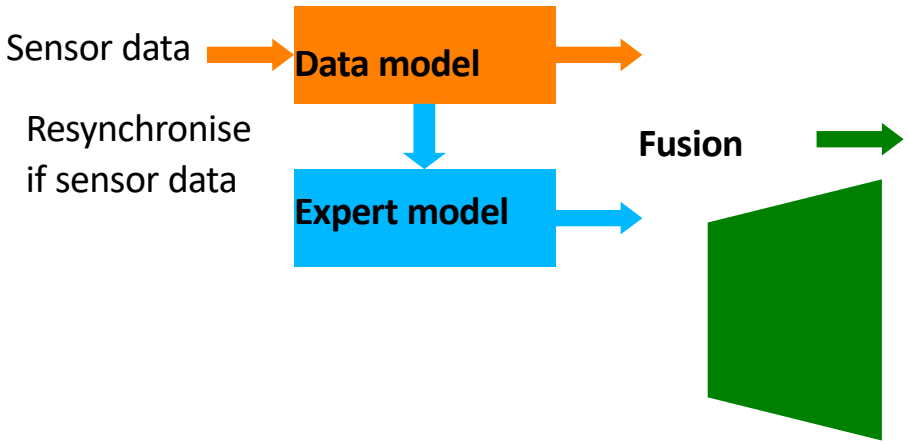
Sleep detection

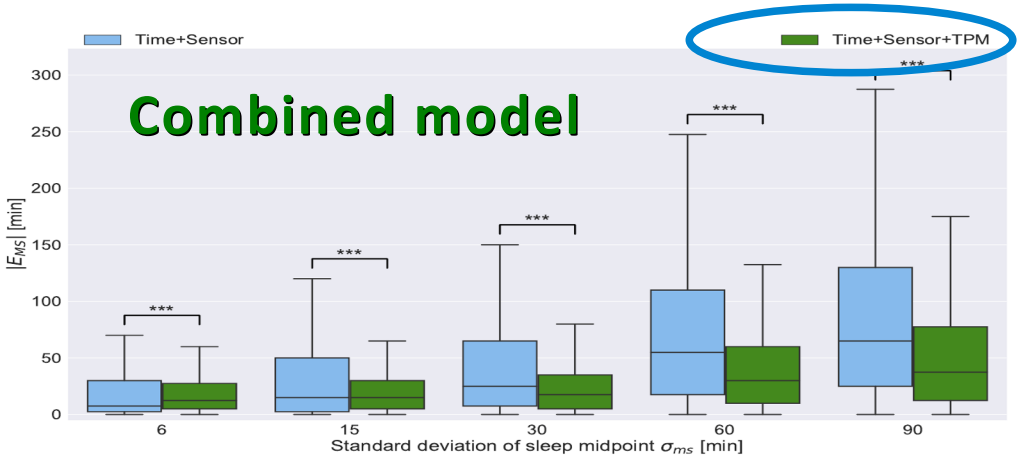
Pure data-based model (boosted decision trees)



Circadian + homeostatic process model (Daan, 1984)

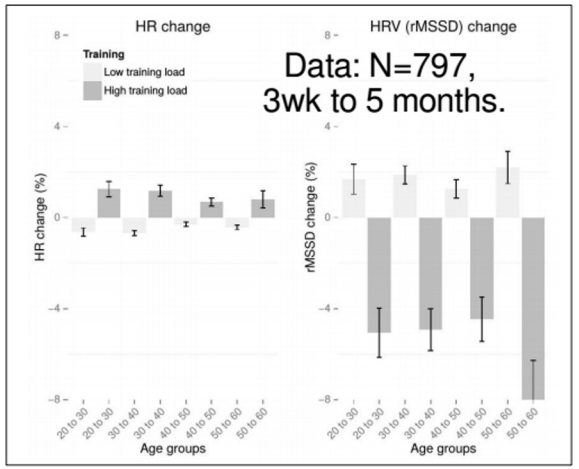




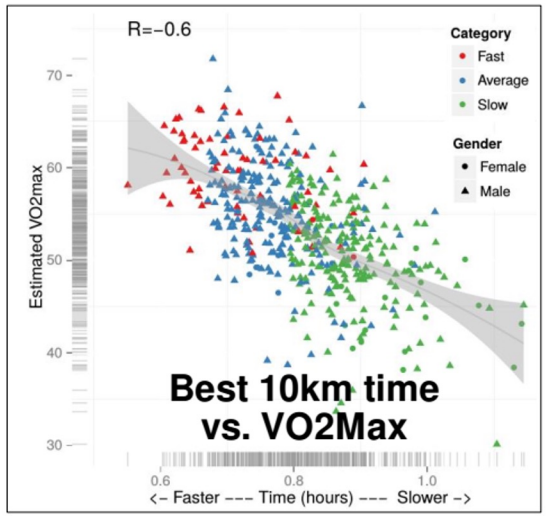


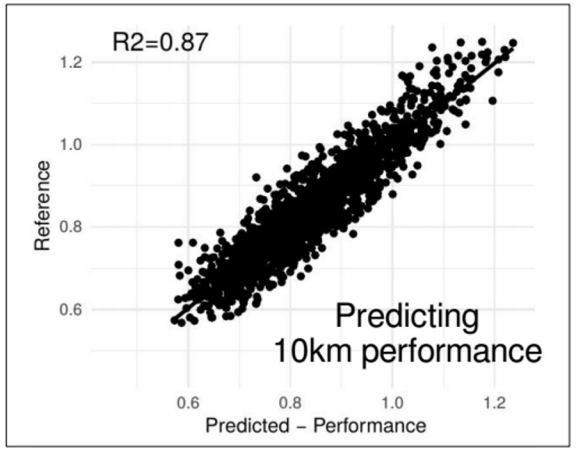
[Wahl et al.,IMWUT, 2018]

HRV4-Training

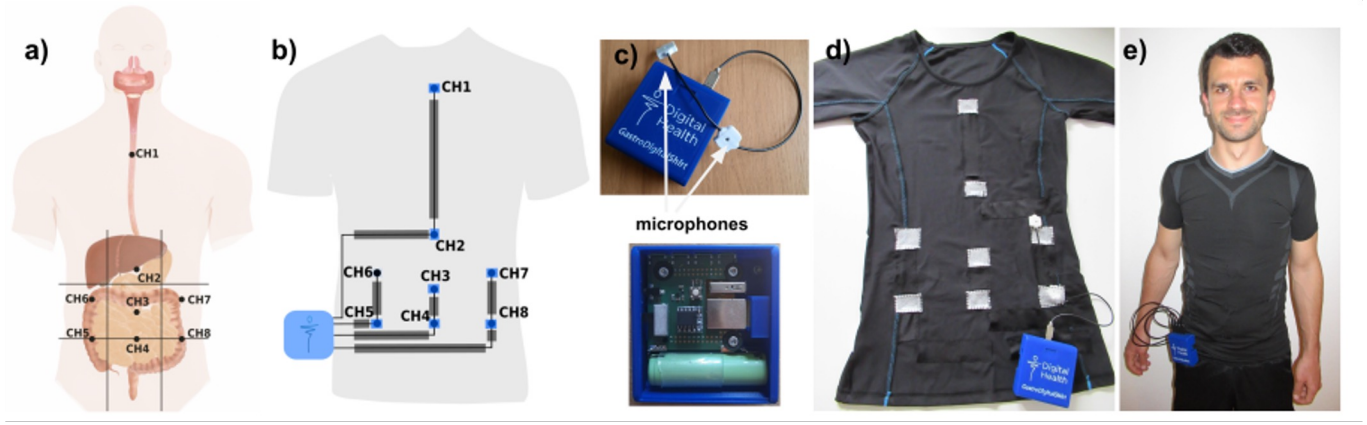


Data:
N=2113
for 2 years.





GastroDigitalShirt



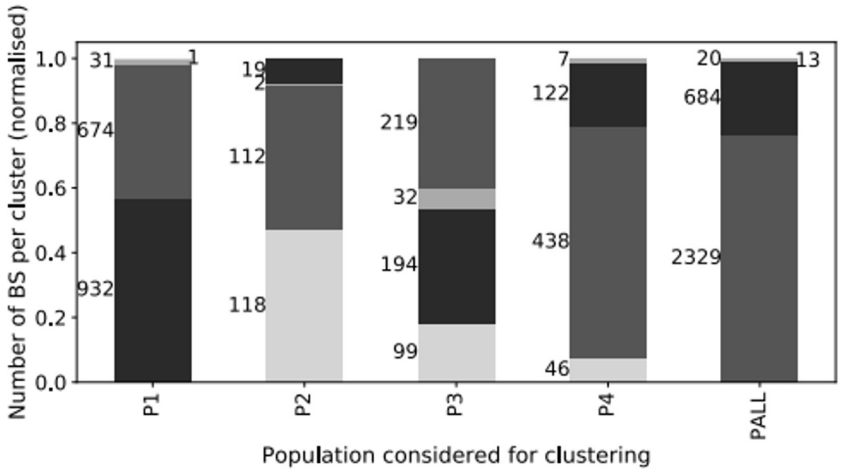


Figure 3: BS instances count per cluster across the different participant datasets (P1, …, P4) and the total joint dataset (ALL). Values next to bars show the BS count per cluster.

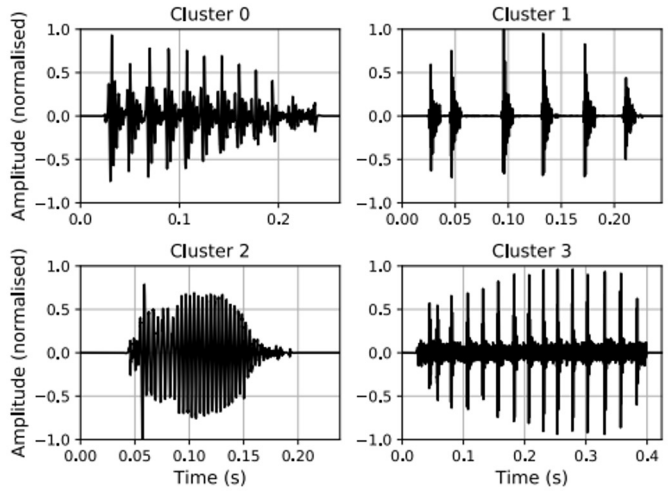
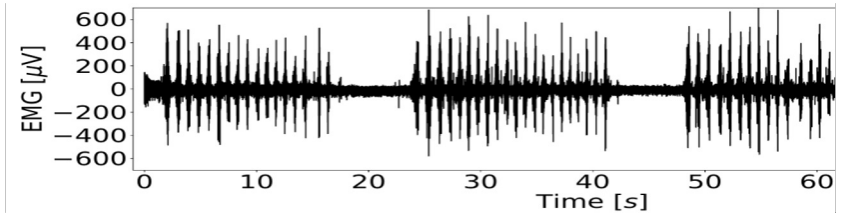
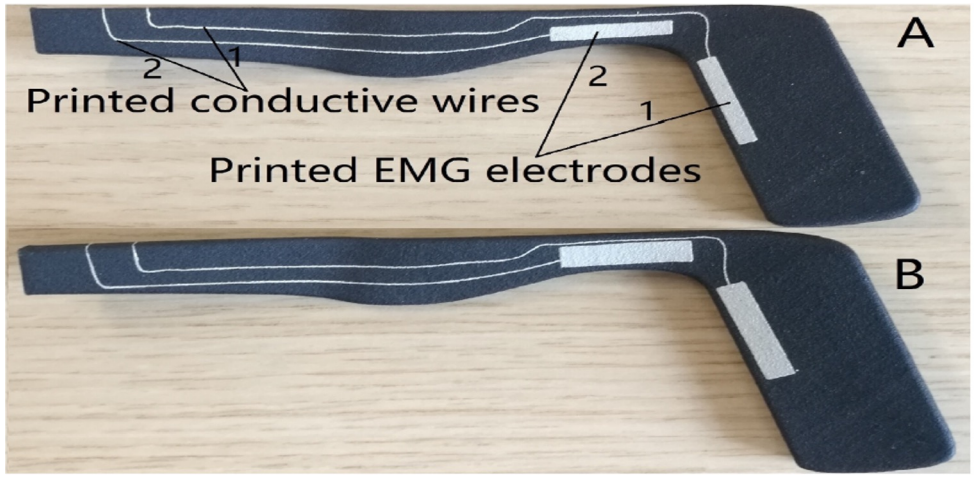


Figure 4: Example BS types separated by the clustering. Top row: most occurring BS clusters (Cluster 0 and 1). Bottom row: BS from Clusters 2 and 3.

Smart Glases

Answer, e.g.: When was food consumed?





[Zhang et al., ISWC, 2018]

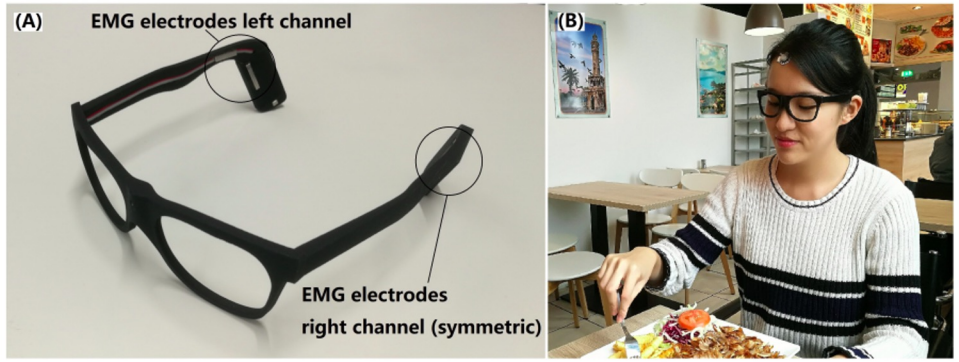
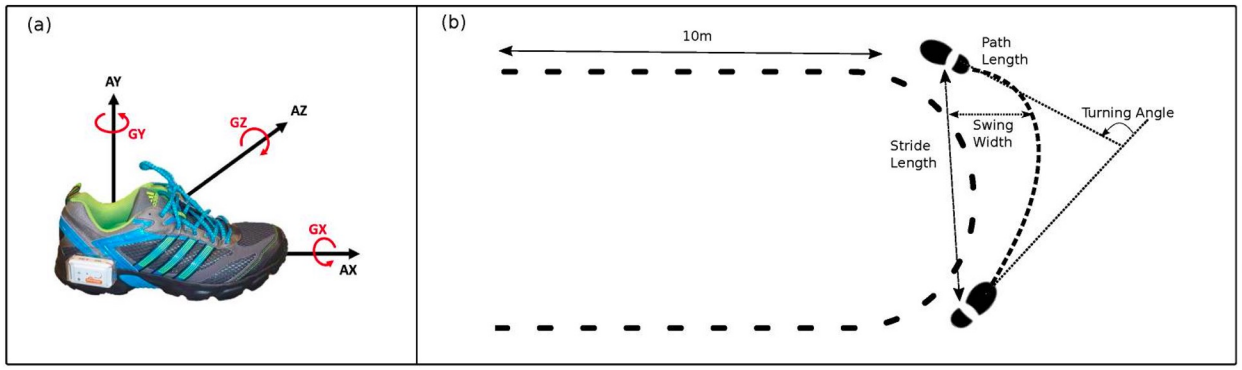


Figure 2. Illustration of the EMG eyeglasses and study: (**A**) Eyeglasses frame with electromyographic (EMG) electrodes symmetrically integrated on the temples. (**B**) Study participant wearing the EMG eyeglasses. Reference EMG electrodes were attached to the skin at the right forehead temporalis muscle position.

[Zhang et al., Sensors 2020]

Smart Shoes



[Ghassemi et al. 2019 Sensors]

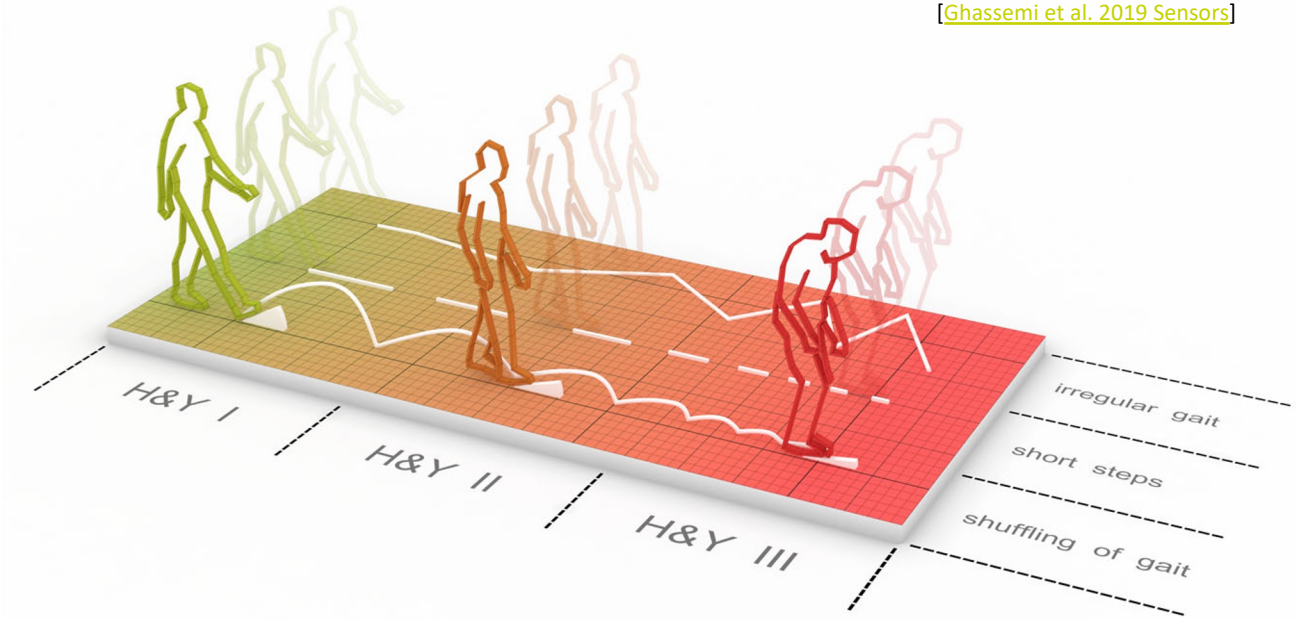


Fig 6. Scheme of measuring individual changes in gait parameters as measured by wearable sensors in PD during the course of the disease.
https://doi.org/10.1371/journal.pone.0183989.g006

Mobile data – Example of time series data processing

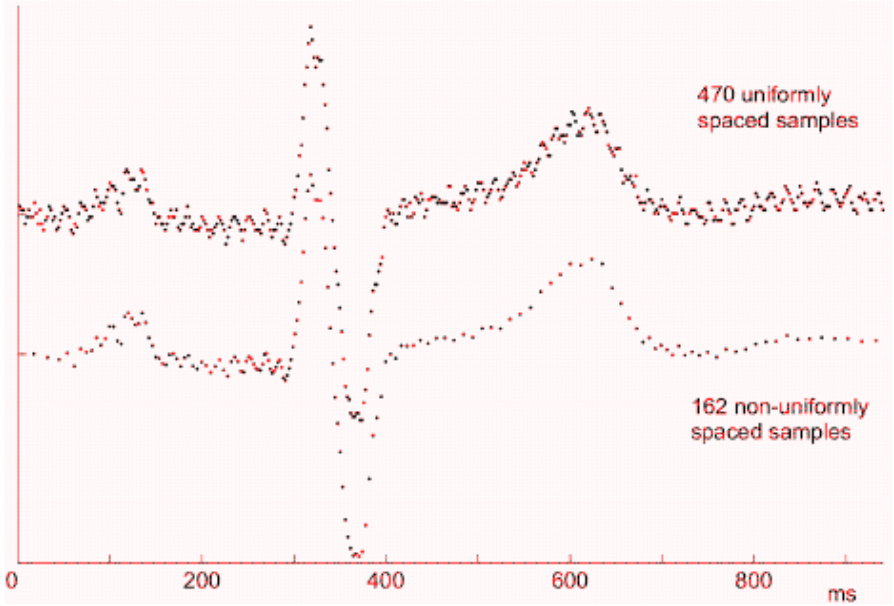
1. Sampling: Select a fraction of the data to reduce volume. Techniques include random, stratified, or systematic sampling.

2. Data Aggregation: Combine data over intervals (e.g., hourly averages) to minimize individual measurements.

3. Feature Engineering: Transform raw data into useful features, like average heart rate over a period or peaks per minute.

4. Data Compression: Use methods like Principal Component Analysis (PCA) to reduce data dimensions.

5. Data Cleaning & Pre-processing: Remove irrelevant or redundant data to streamline the dataset.



Piotr Augustyniak, 2005

Mobile data summary

Types of ambient data:

- Physical quantities: e.g., body temperature, PPM
- Activity data: e.g., running, cycling
- User state data: e.g., stress level, arousal

Characteristics:

- Measured on the body
- Uses an on-body computing device as intermediary and data preprocessor
- Continuous measurement

Ambient data

Ambient data

Photo by Stephan Bechert on Unsplash

Environmental data

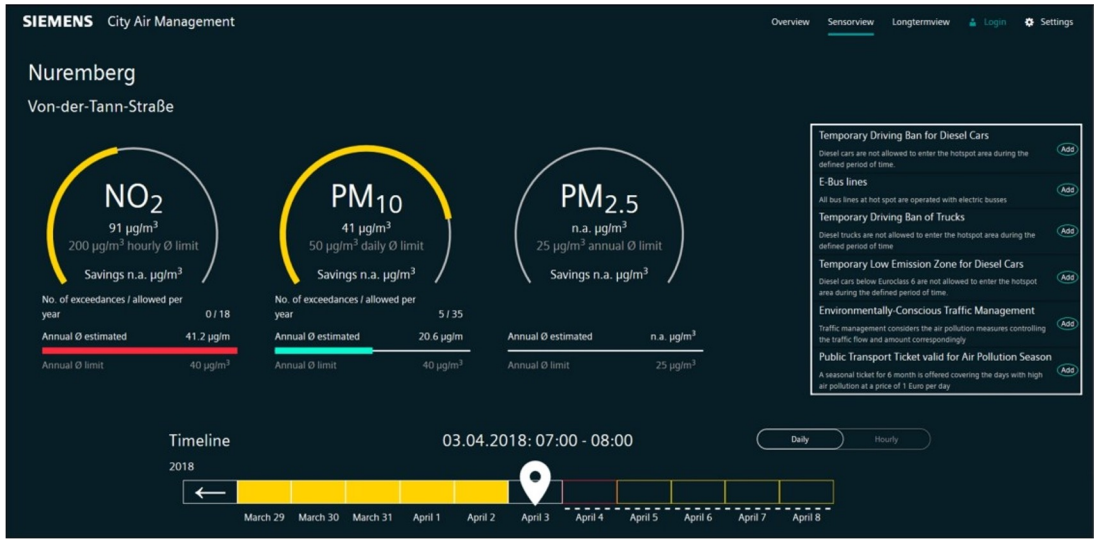
WHO Ambient air pollution

An estimated 4.2 million premature deaths globally are linked to ambient air pollution, mainly from heart disease, stroke, chronic obstructive pulmonary disease, lung cancer, and acute respiratory infections in children. Worldwide ambient air pollution accounts for:

- *29% of all deaths and disease from lung cancer*
- *17% of all deaths and disease from acute lower respiratory infection*
- *24% of all deaths from stroke*
- *25% of all deaths and disease from ischaemic heart disease*
- *43% of all deaths and disease from chronic obstructive pulmonary disease*

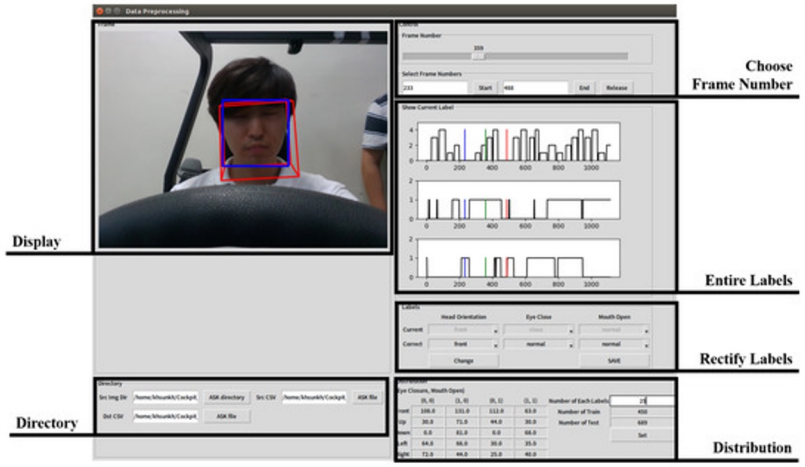
Photo by Thomas Winkler on Unsplash

Photo by Lloyd Dirks on Unsplash



Siemens - Artificial-intelligence improves air quality

User state



[Kim et al. Sensors 2019]

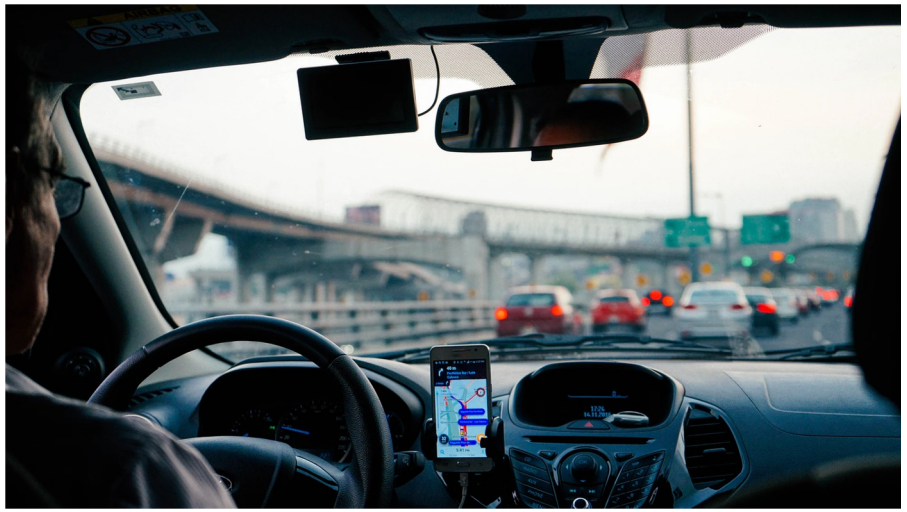
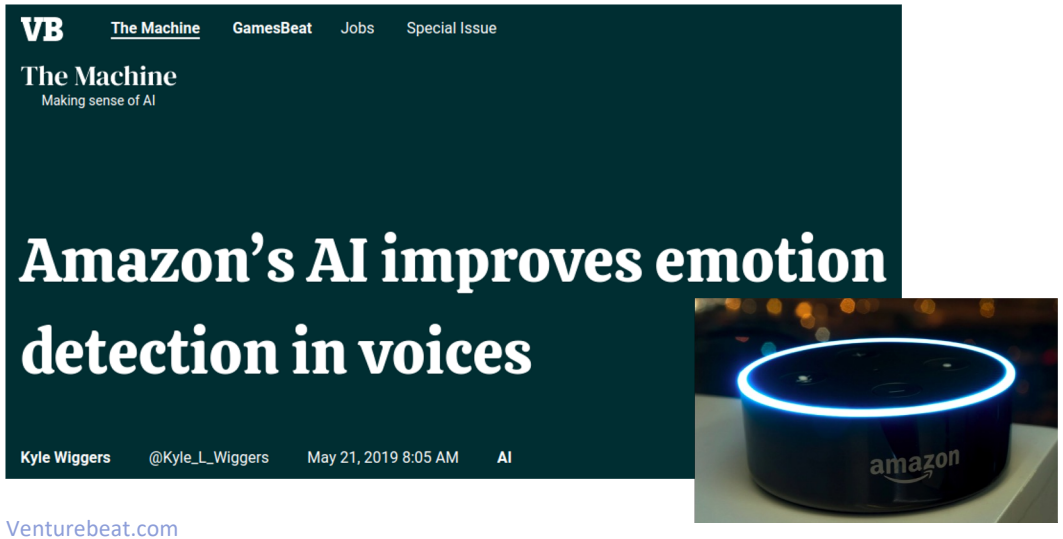


Photo by Dan Gold on Unsplash



Venturebeat.com

User behaviours

Product Hunt TOP PRODUCTS LIVE BOOKS COMPANY ARCHIVE NEWSLETTER

Must-Have Smart Kitchen Gadgets that Will Help You Cook Better and Eat Healthier

Product Hunt Follow
Apr 19, 2017 · 10 min read

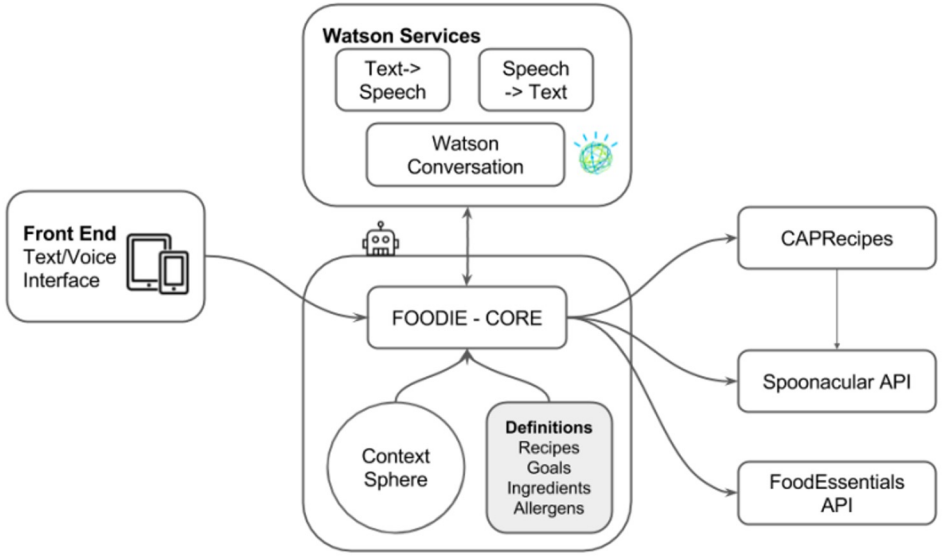


Figure 1: Foodie's architecture and dependencies

Foodie Fooderson A Conversational Agent for the Smart Kitchen

Ambient data summary

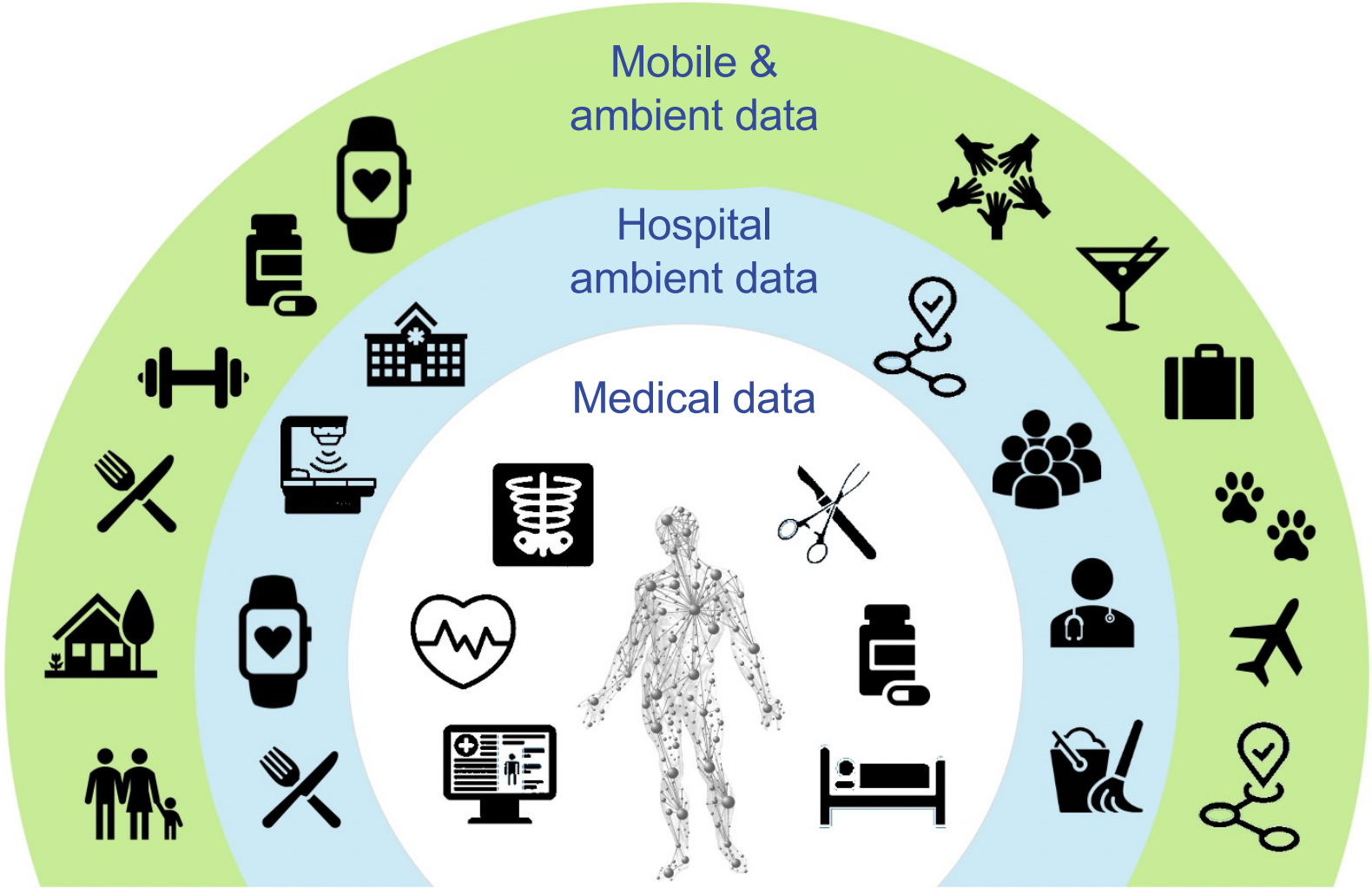
Types of ambient data:

- Physical quantities: e.g., temperature, humidity, pollution and oxygen
- Activity data: e.g., cooking and working at the desk.
- Behavioural data: e.g., eating schedules and meal types
- User state data: e.g., stress level

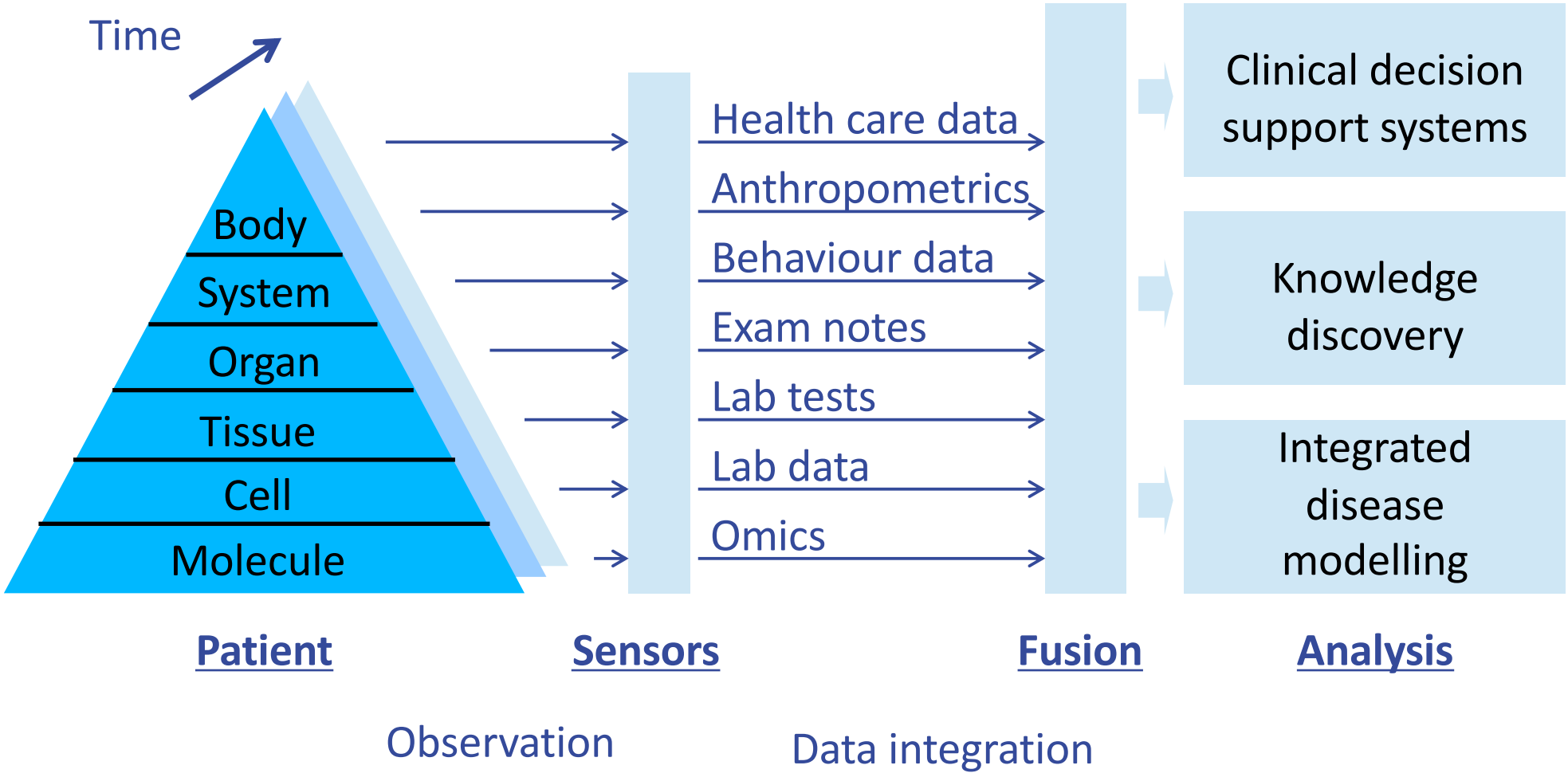
Characteristics:

- Measured on building or field mounted sensors
- Uses IoT devices and cloud infrastructure
- Continuous measurement

Data collection for health and workflow prediction



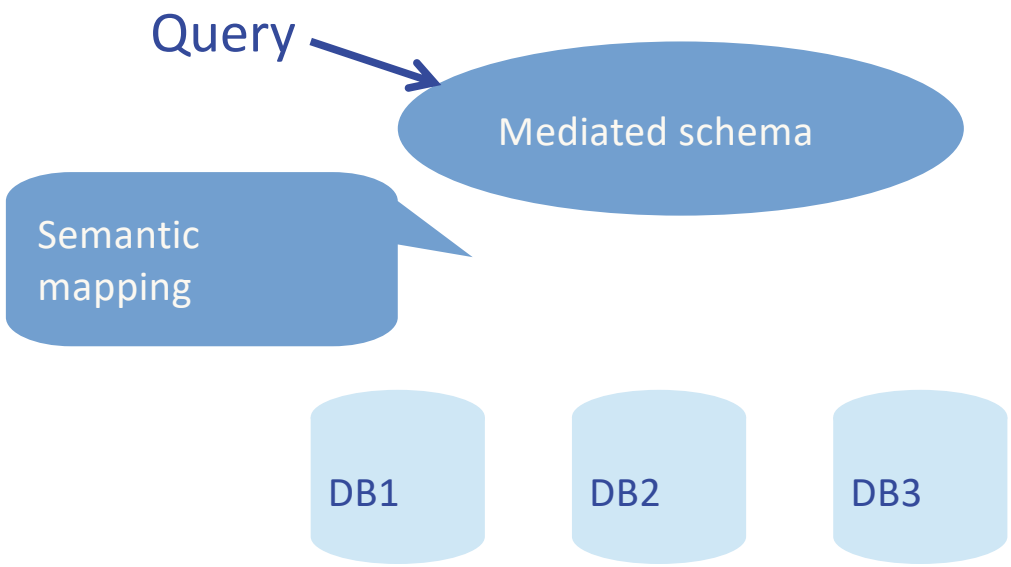
Conceptual multi-model approach



Integration principles

Data is created independently by stakeholders, e.g. in EMRs. Different systems, different database (DB) schemata, have limited interfaces to data, and are controlled by many people.

Goal of data integration: tie together different sources under common schema
→ a higher-level abstraction



Independence of:

- source & location
- data model, syntax
- semantic variations

Exemplary sources for medical data

https://archive.ics.uci.edu
https://www.kaggle.com

Specifically for imaging:
https://medpix.nlm.nih.gov/home
https://imaging.cancer.gov
https://www.oasis-brains.org

Specifically for time series data:
https://physionet.org
https://guides.library.ucla.edu/psychology/data
…

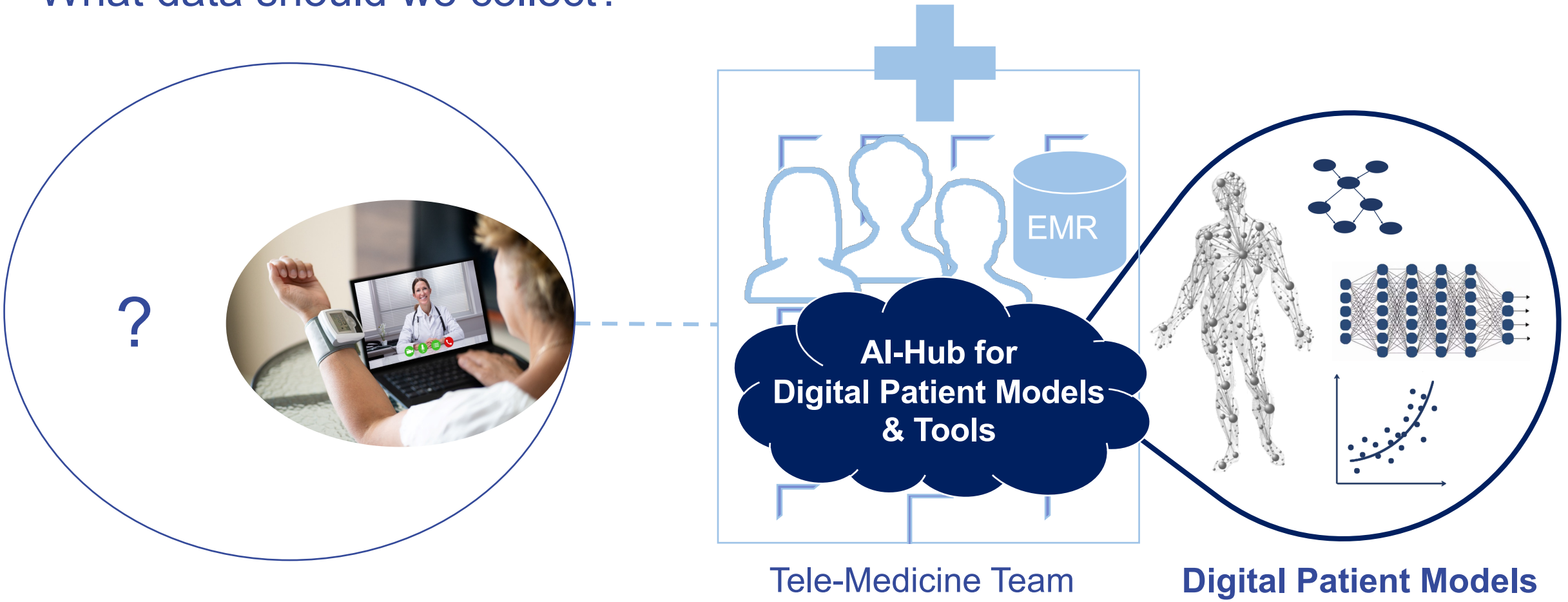
Activation!

11. Why does mobile data leads to a paradigm shift in healthcare?
12. What does mobile data has in common?
13. What challenge brings mobile data in healthcare?
14. Does “smart” gym equipment produces ambient data or mobile data?
15. How do we classify internet activity?

pingo.coactum.de/815896

Activation! | Tele-Aftercare for Kidney Transplant Patients

What data should we collect?



SmartNTx

smart NTx

Deutsches Forschungszentrum für Künstliche Intelligenz GmbH

Uniklinikum Erlangen

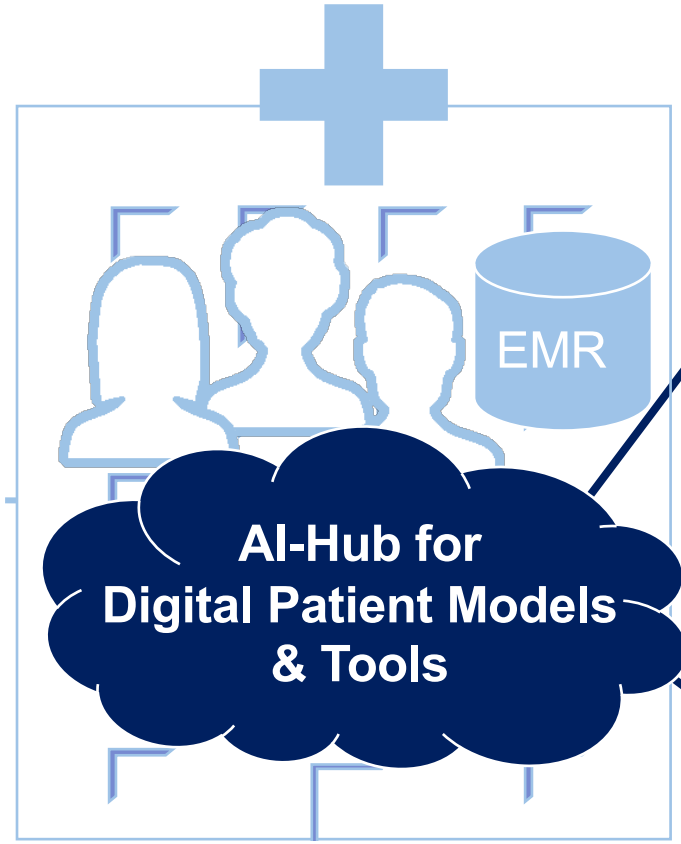
CHARITÉ UNIVERSITÄTSMEDIZIN BERLIN

Universitätsklinikum Essen

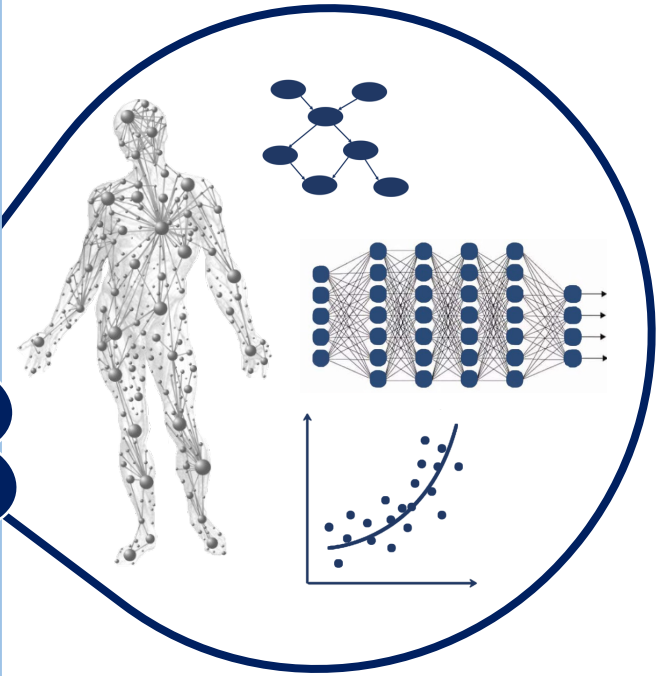
Hahn Schickard

Gemeinsamer Bundesausschuss Innovationsausschuss

Smart Home Devices



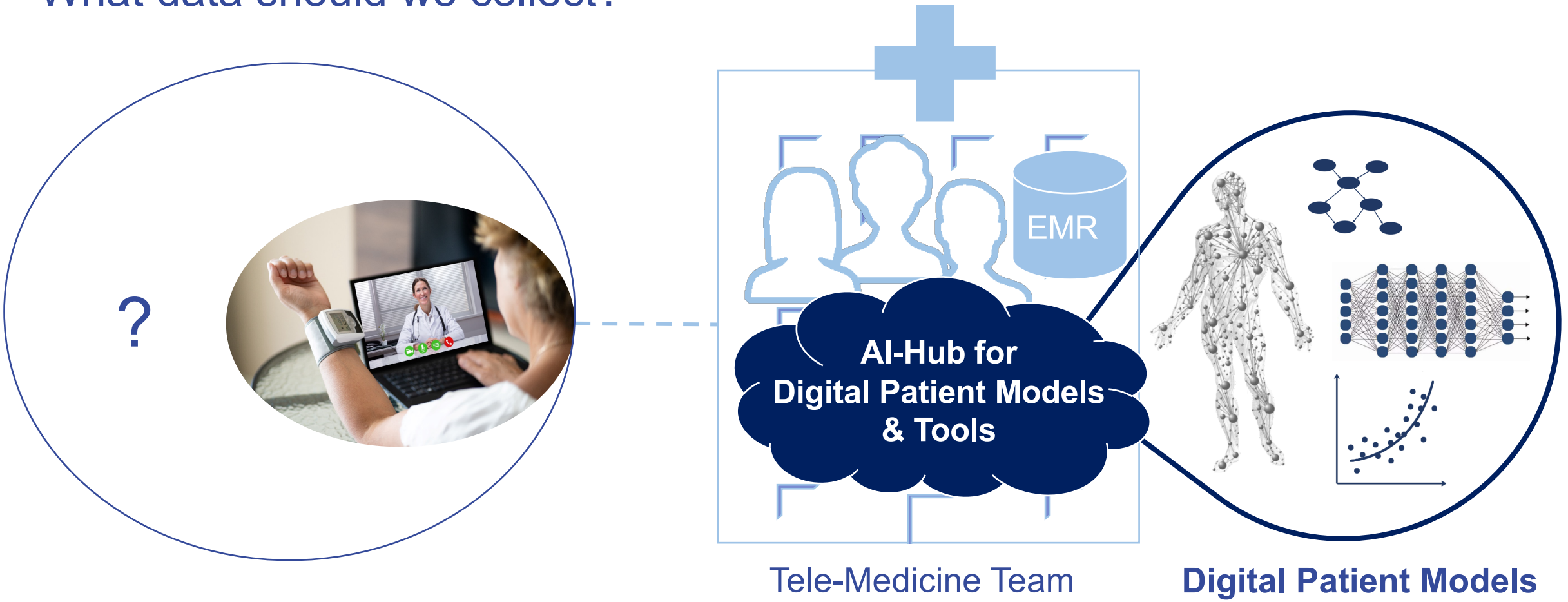
Tele-Medicine Team



Digital Patient Models

Activation! | Tele-Aftercare for Gastrointestinal Patients

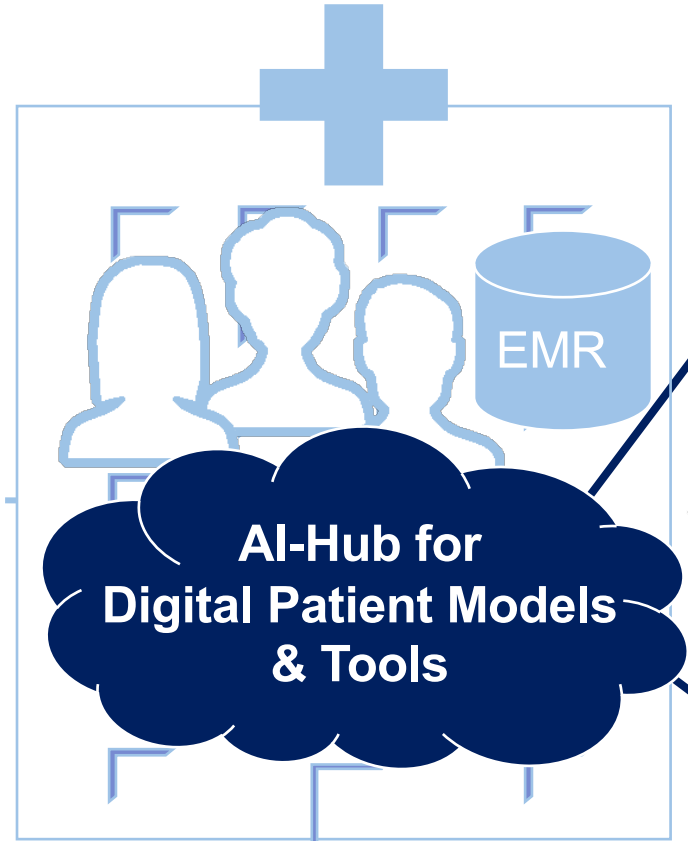
What data should we collect?



KI-COMPASS

Bundesministerium für Forschung, Technologie und Raumfahrt

Smart Home Devices



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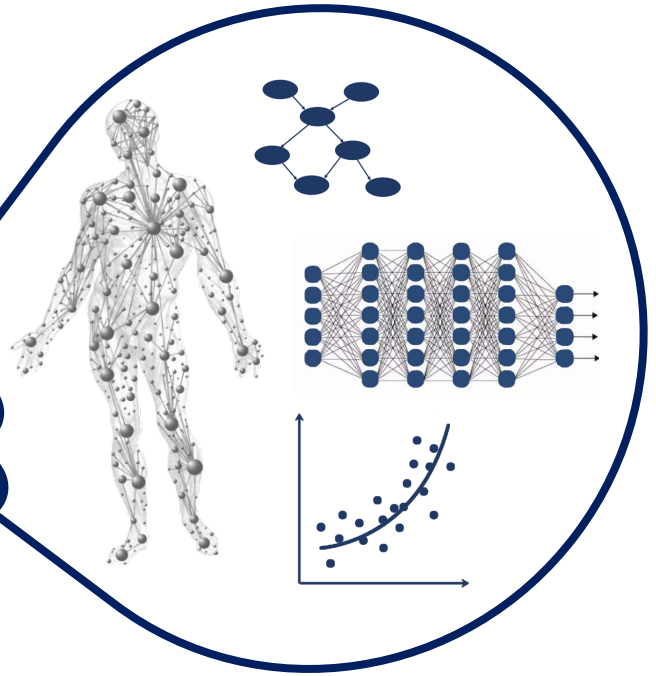


Digital Patient Models

Balancing Data Collection and Value: Different Viewpoints

Balancing Data Collection and Value: Different Viewpoints

Patient

Clinicians

Insurance

Regulatory & Government

Researcher & Pharma

Balancing Data Collection and Value: Different Viewpoints

Clinicians

- **Value:**
  - Enhanced diagnostic capabilities
  - Personalized treatment planning
  - Monitoring treatment response
  - Clinical research and evidence-based practice
- **Concerns:**
  - Data reliability and accuracy
  - Data overload vs. actionable insights
  - Integration with current systems
  - Patient privacy and consent issues

Regulatory & Government

Researcher & Pharma

Ref: Mandl KD et al. "The SMART Platform: Early Experience Enabling Substitutable Applications for Electronic Health Records." J Am Med Inform Assoc. 2012.

Balancing Data Collection and Value: Different Viewpoints

Patient

- **Value:**
  - Informed decision-making
  - Empowerment and self-management of health
  - Seamless communication with healthcare providers
  - Access to new treatments and interventions
- **Concerns:**
  - Privacy and data misuse
  - Over-monitoring leading to anxiety
  - Understanding complex medical data
  - Digital literacy barriers

Regulatory & Government

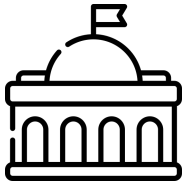
Researcher & Pharma

Ref: Patel MS and Asch DA. "The Role of Patient Engagement in Addressing Parents' Perceptions About Immunizations." JAMA. 2017.

Balancing Data Collection and Value: Different Viewpoints

Insurance

- **Value:**
  - Risk assessment and stratification
  - Fraud detection
  - Cost-saving by early disease detection and intervention
  - Tailored insurance plans based on individual data
- **Concerns:**
  - Discrimination based on data insights
  - Data reliability and accuracy
  - Ethical implications of data use
  - Regulation and compliance issues



Regulatory & Government

Researcher & Pharma

Ref: Obermeyer Z et al. "Dissecting racial bias in an algorithm used to manage the health of populations." Science. 2019.

Balancing Data Collection and Value: Different Viewpoints

Researcher & Pharma

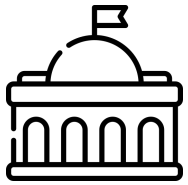
- **Value:**
  - Population health monitoring
  - Policymaking based on real-world data
  - Ensuring healthcare standards and quality
  - Responding to public health crises
- **Concerns:**
  - Ethical oversight of data collection and use
  - Balancing individual rights with public health needs
  - Ensuring equitable access and representation in data
  - Avoiding potential misuse by authorities

Regulatory & Government

Researcher & Pharma

Ref: Cohen IG and Mello MM. "Big Data, Big Tech, and Protecting Patient Privacy." JAMA. 2019.

Balancing Data Collection and Value: Different Viewpoints



Regulatory & Government

- **Value:**
  - Accelerated drug discovery
  - Real-world evidence studies
  - Patient recruitment for clinical trials
  - Drug safety monitoring
- **Concerns:**
  - Ensuring data quality for research
  - Patient consent for data use in research
  - Avoiding biases in datasets
  - Ethical considerations in data-driven research

Regulatory & Government

Researcher & Pharma

Ref: Platt R, et al. "The New Sentinel Network — Improving the Evidence of Medical-Product Safety." N Engl J Med. 2009.

Activation!

Who’s viewpoint is most important?

Values vs. concerns - How to solve it?

What could be a good rule of thumb for collecting patient data?

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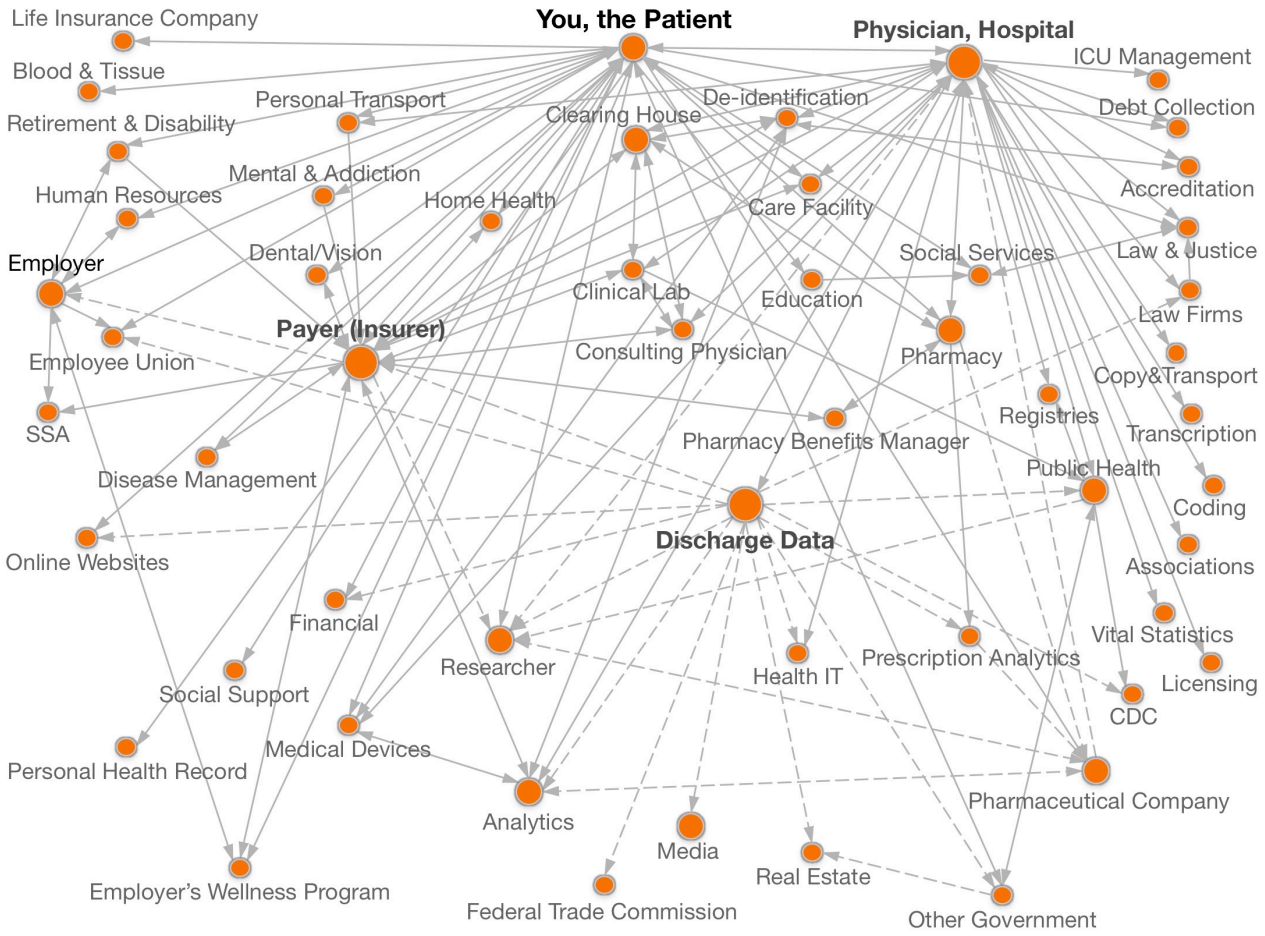
Mechanisms to increase data privacy and security

Data privacy and security



Mapping of places where patient data goes by https://thedatamap.org

Example for the impact of a data breach

MEDICAL INFORMATICS ENGINEERING

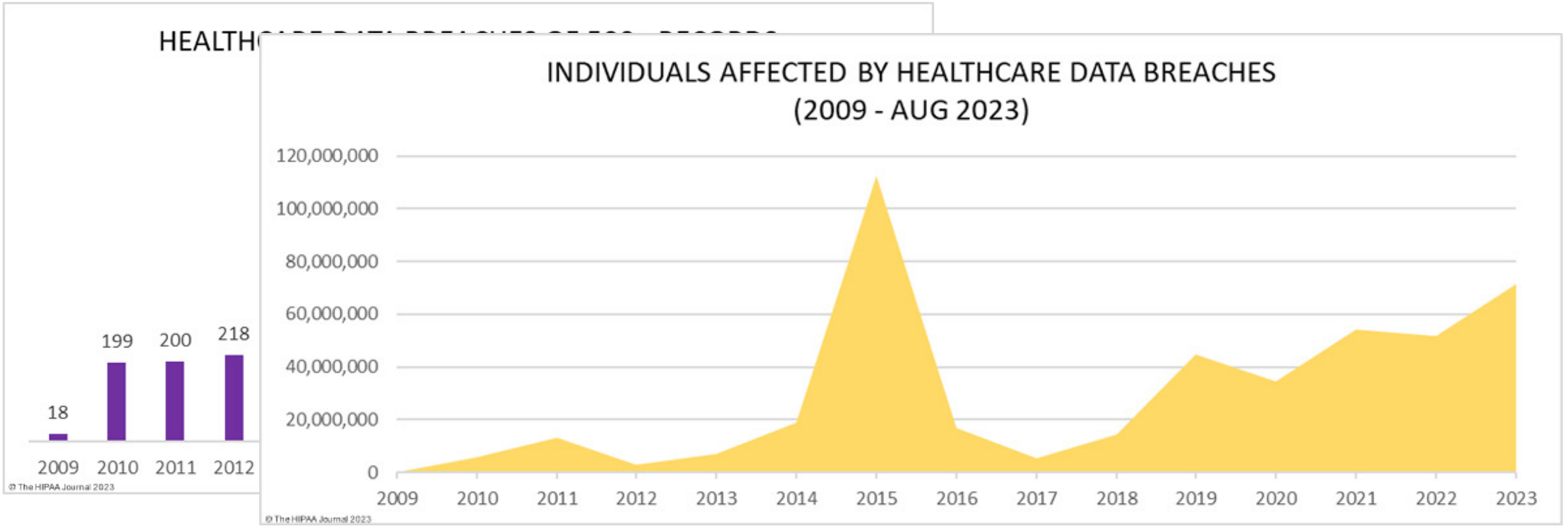
Date: July 2015
Impact: 3.9 million patients

"Cybercriminals accessed one of MIE's servers by using a compromised username and password and maintained undetected access for 19 days."

https://www.upguard.com/blog/biggest-data-breaches-in-healthcare

Data privacy and security

Statistical representation of healthcare data breaches by HIPPA Journal



https://www.hipaajournal.com/healthcare-data-breach-statistics/

Mechanisms to increase data privacy and security

- Data Encryption
- Data Access Control
- Regular Auditing and Monitoring
- Data Anonymization and De-identification
- Secure Data Storage
- Network Security
- Security Training and Awareness
- Regular Software Updates and Patching
- Compliance with Standards and Regulations
- Risk Management

Mechanisms to increase data privacy and security

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Technical mechanisms

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Administrative Mechanisms

Mechanisms to increase data privacy and security

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Legal and Regulatory Mechanisms

Mechanisms to increase data privacy and security

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e.g.,
General encryption in the storage or
SSL/TLS for web traffic

Mechanisms to increase data privacy and security

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e.g.,
Role-Based Access Control: Assigns access based on roles within an organization.
Access Control Lists (ACLs): Specify which users/groups can access what data.
Data Use Agreements (DUAs): Contracts that specify how data can be used, ensuring third parties handle data responsibly.

Mechanisms to increase data privacy and security

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e.g.,

Logging: Recording data access and modification for review.

Intrusion Detection Systems: Monitors system for malicious activities or policy violations.

Incident Response Plan: Prepared strategies to address and manage a data breach.

Mechanisms to increase data privacy and security

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e.g.,
Pseudomization: Replacing names by random IDs.

Tokenization: Replacing sensitive data with non-sensitive placeholders.
Generalization: Reducing the granularity of the data, e.g., changing exact age to an age range.
Data Masking: Concealing certain data within a database, rendering the data anonymous.

Mechanisms to increase data privacy and security

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e.g.,
Trusted Platform Module (TPM): A specialized chip on a device to store RSA encryption keys specific to the host system.
Hardware Security Module (HSM): A physical computing device that safeguards digital keys for strong authentication.

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- Risk Management

e.g.,

Firewalls: Filters incoming/outgoing traffic based on a set of security rules.

Virtual Private Networks (VPNs): Creates a secure connection over a less-secure network.

Network Segmentation: Dividing a computer network into subnetworks to improve performance and security.

Mechanisms to increase data privacy and security

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- Risk Management

e.g.,
Regular training sessions for employees.
Phishing tests to ensure employees can identify malicious attempts.

Mechanisms to increase data privacy and security

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- Security Training and Awareness
- **Regular Software Updates and Patching**
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- Risk Management

e.g.,
Ensuring all software, especially Electronic Health Record (EHR) systems, are up to date.
Quick patching of known vulnerabilities.

Mechanisms to increase data privacy and security

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- Regular Software Updates and Patching
- **Compliance with Standards and Regulations**
- Risk Management

e.g.,

Health Insurance Portability and Accountability Act (HIPAA): U.S. legislation that provides data privacy and security provisions to safeguard medical information.

General Data Protection Regulation (GDPR): EU legislation on data protection and privacy.

ISO/IEC 27001: International standard for information security management systems.

Mechanisms to increase data privacy and security

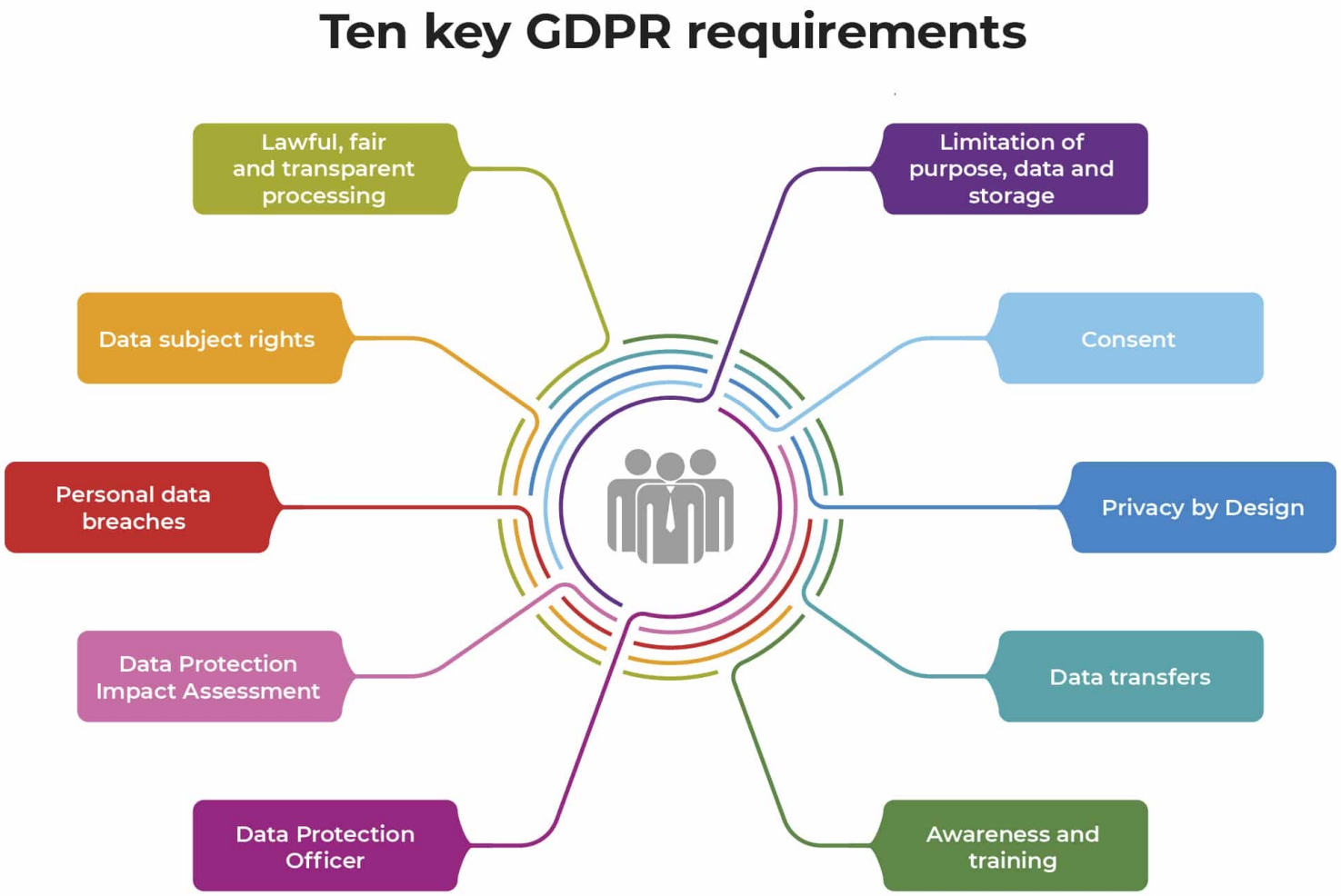


Image by https://advisera.com

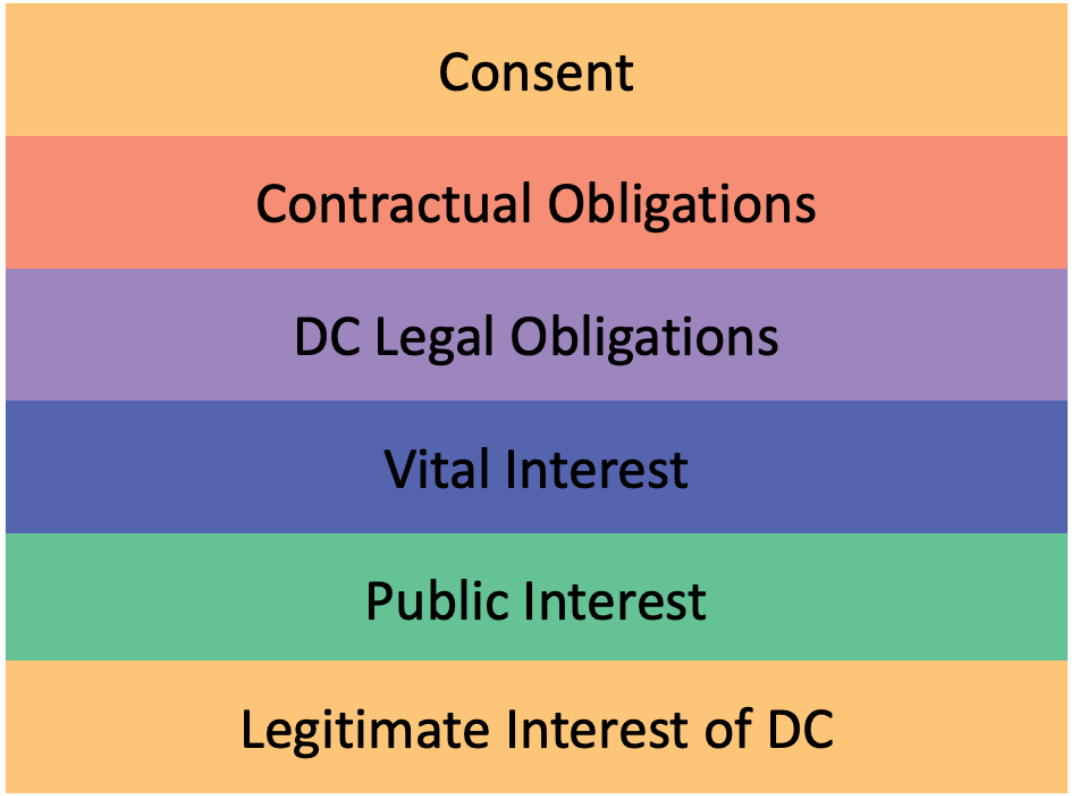
Data Subject = Patient

Data Controller = Deteriming puprose and means of processing

Data Processor = Processes data on behalf of data controller

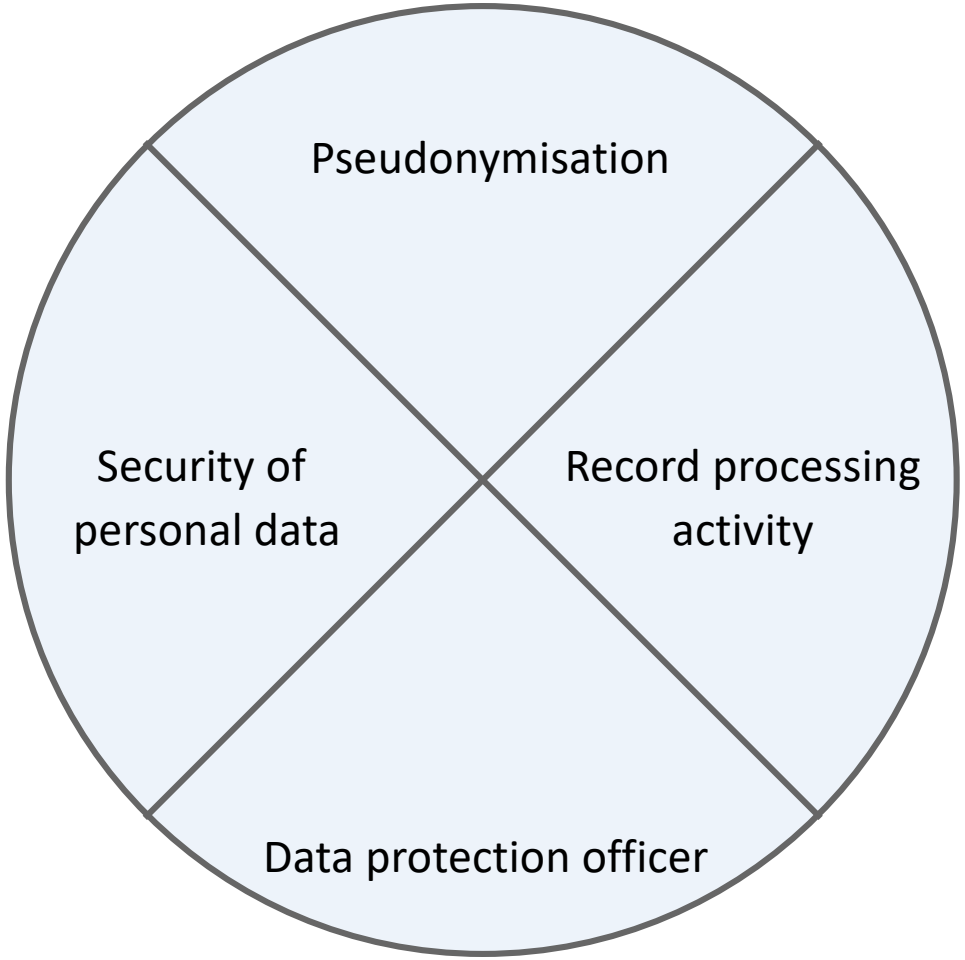
Mechanisms to increase data privacy and security

Personal data might **only** be processed under the following legal basis.



Controller and Processor

Data protection by design



Mechanisms to increase data privacy and security

ISO/IEC 27001: International standard for information security management systems.

Image by www.imperva.com

Mechanisms to increase data privacy and security

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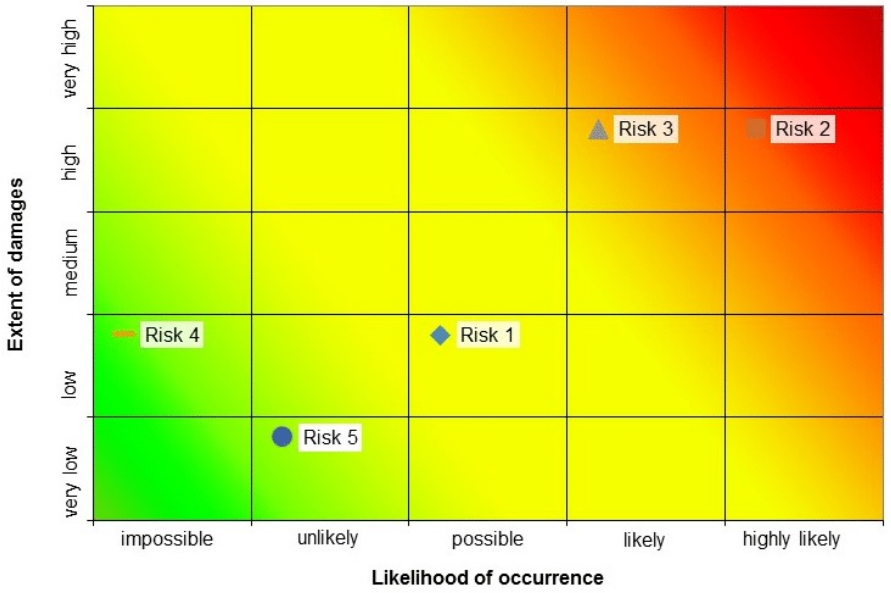
- **Risk Management**

e.g.,
Ensuring third-party vendors adhere to security standards.
Regular security assessments.



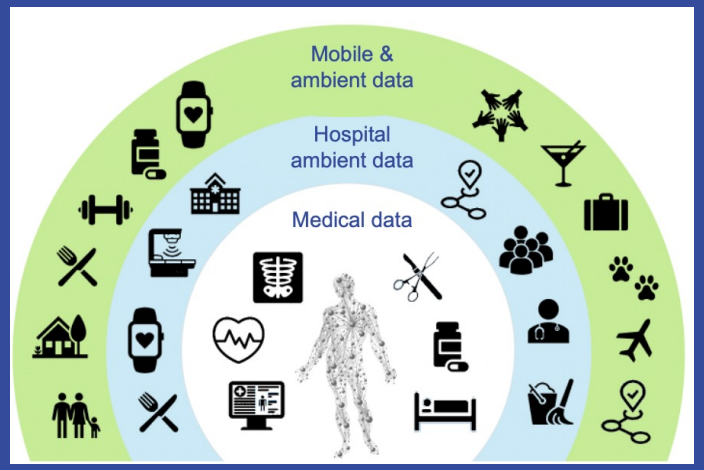
Additional material

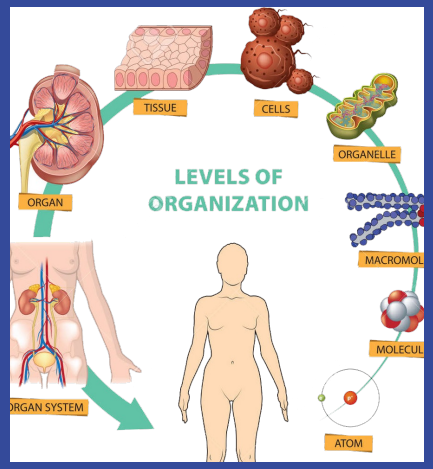
What is GDPR?	https://www.youtube.com/watch?v=Assdm6fIHlE
GDPR (Wikipedia)	https://en.wikipedia.org/wiki/General_Data_Protection_Regulation
GDPR Documents	https://eur-lex.europa.eu/eli/reg/2016/679/oj

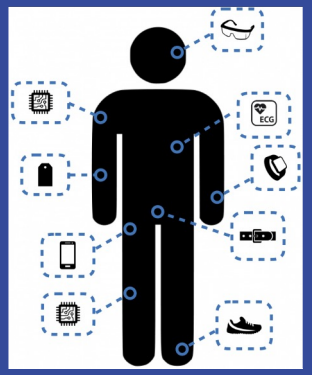
Quiz

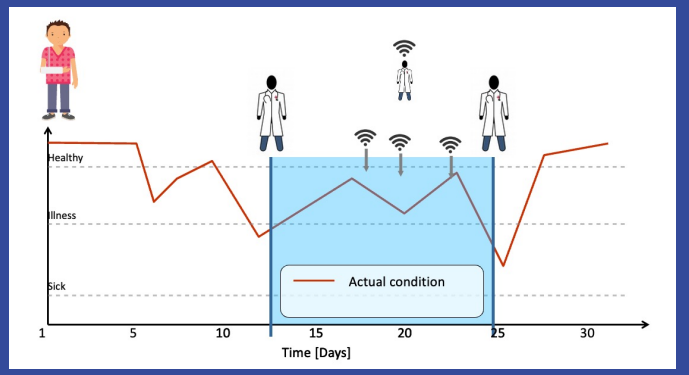
1. Where is health data collected?
2. What is the difference between genomics, proteomics, metabolomics and exposomics?
3. Name typical medical data types.
4. Name and explain methods for data preprocessing.
5. What is ambient data and how can we collect it?
6. Why does the continuous collection of vital signs lead to a paradigm shift in clinical routine?
7. What is evidence-based medicine?
8. Explain the principle of converting data to wisdom.
9. What is the difference between EHR, EMR, and MR?
10. Who is affected by data security and what concerns are there about data use?
11. Why is it important to balance between data collection and value?
12. Which mechanisms can increase data security and data protection?

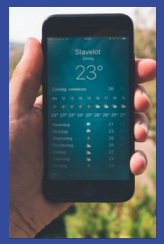
End – Questions?

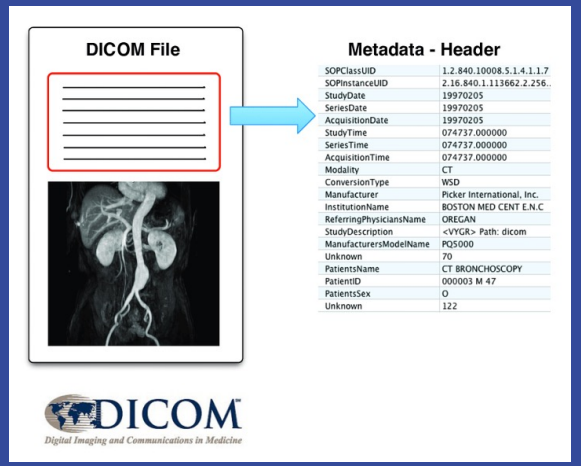


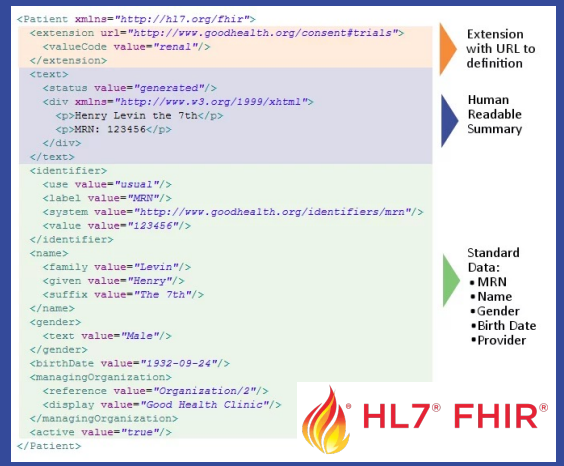


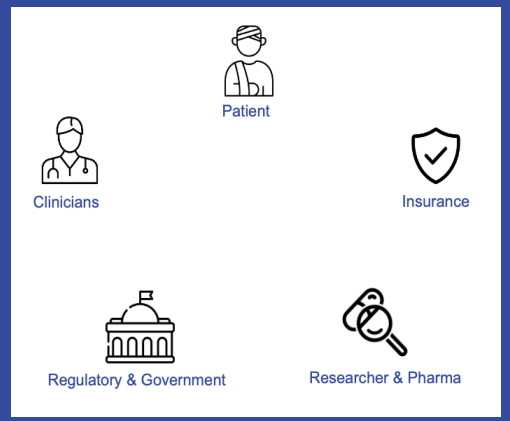


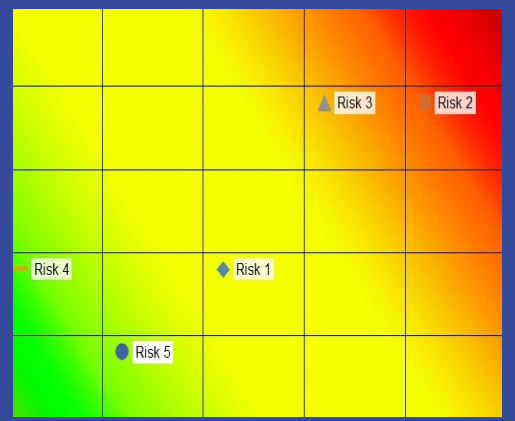












IES Website (WordPress)

Student Research Assistant
10 h / week

Intelligent Embedded Systems Lab

Hiring: Student Research Assistant (10 h/week) – IES Website

- WordPress + HTML/CSS; **JS ideal**
- Initiate improvements, **migrate** content, **refine** UX
- Build/integrate **literature database plug-in**
- Flexible, impactful HiWi role

Apply:koeberg@informatik.uni-freiburg.de | Subject: “HiWi – IES Website”

Start: ASAP | Location: Hybrid | Pay: Univ. HiWi rates